nature portfolio

Peer Review File

Cell-Free DNA Testing for the Detection and Prognosis Prediction of Pancreatic Cancer

Corresponding Author: Professor Lei Chen

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Attachments originally included by the reviewers as part of their assessment can be found at the end of this file.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
Wu et al. reported cfDNA-based models for pancreatic cancer diagnosis and prognosis prediction. The topic is very important, however, the manuscript is terribly written, very difficult to follow, and many details are hidden. Since both the data and codes are unavailable, it is impossible to evaluate the reliability of the results.

Detailed comments:
1. There is no need to have a Suppl. Method, the authors should move the related content to main text.

2. for NF, the methods part "Nucleosome footprint model was built by bedtools to calculate the reads distribution and remove uncovered genes at the TSS +/- 2.5kb region". This is far from complete, and I do not understand how to calculate NF. Is it a value, or a vector of many values?

3. The authors used 20kb bins to screen for CNVs, is it too small that may cause high variations?

4. To calculate CNVs, a group of healthy subjects are used. How many? Are these samples also used in diagnosis, either training or testing?

5. "To obtain the best surveillance model". I am confused, as the logistic regression is designed for diagnosis?

6. Their are lots of formula, how did not determine the parameters? e.g., 2.13, 3.26, 2.85 in Logistic Score?

7. For SVM, is it just used for feature selection? And you used its model to get calculate an intermediate feature, then arteficially mixed it with the features used in SVM again to build a PCM score. This is very confusing and makes no sense. Why not build a diagnostic model using the features directly?

8. There are duplicated sentences in "Patients and cohorts".

9. "We also collected serum from patients with other types of cancer". This is funny, as your model is built using plasma, but you validate it using serum?

10. Many sentences in Discussion should be put into Introduction, as they are background information, and the authors did not discuss the results using them.

11. What is the advance over the previous works on pancreatic cancer (ref. 26-28). This part actually should be discussed in detail.

12. "next generation sequencing (NGS) technology to acquire plasma cfDNA end motif [29]". Citation 29 is "Branco MR, Ficz G, Reik W. Uncovering the role of 5-hydroxymethylcytosine in the epigenome. Nat Rev Genet 2011; 13: 7-13", why cite this one?

13. Would the performance improve by integrating the protein biomarkers?

Reviewer #2

(Remarks to the Author)
This study constructed models for early detection and prognosis of pancreatic cancer using features of cell-free DNA (cfDNA) in plasma and validated these models by using two external cohorts. Early detection of pancreatic cancer is a clinically important unmet need and this study is timeline. However, insufficient documentation and supporting results significantly hinder transparency and reproducibility of the research. Another concern is that the so-called external cohorts are not genuinely external since the healthy samples assigned as external cohort were processed and collected in the same way as training cohort. Therefore, this might overestimate generalizability of their model, as it is not truly unseen data. Furthermore, the documentation of this study requires substantial improvement. Most of computational procedures are mostly unmentioned in the methods section. Additionally, some computational code/file they distributed is encrypted and cannot be opened.

Major comment
1. Limit of detection of the model
It is well known that tumor content in cfDNA significantly affects the detection limit of cancer and any other cfDNA applications. Previous research using cfDNA have already demonstrated that cancer detection in samples with a high amount or proportion of circulating tumor DNA (ctDNA) is relatively easy. I recommend the authors testimate ctDNA content using other tools (e.g., ichorCNA [PMID: 29109393]) and report an approximate limit of detection in terms of ctDNA content.
2. Among the main results, only figures 4b,c support the possibility of early detection by the model. It would be great to evaluate each cohort separately (testing, external 1 and 2) in the analysis in figures 4b,c.
3. Rather than simply providing a KM plot, it would be helpful to show how many high-risk patients can be accurately identified in the external cohort using the PCP score and defined threshold (i.e., 0.3127).
4. The healthy samples in training and external validation cohort are collected in the same way. It seems they just randomly distributed to training/testing/external validation cohorts. This is a problematic design that could allow leakage of information into the external validation cohorots. For instance, if all of the healthy samples vary is some way that is NOT generalizable and not due to the absence of cancer, the model may seem to perform well on external validation, but fail when applied to truly “unseen” cohorts.
5. The method section could use more detail. While certain analysis and data are mentioned in the results, they are completely omitted in the method section and supplementary methods. Particularly, Python, R and Perl code can be found in their GitHub repository, but only Python programs are mentioned in the manuscript.
At the very least, the following items must be clearly explained:
• line 150: procedure of GC correction
• line 154: The rationale of the Z-score threshold. Is it a threshold which healthy samples do not present CNV?
• Line 160: Why are CNV scores for tumor suppressor genes and oncogenes are calculated separately?
• Line 176: What is the rationale for nucleosome footprint, motif and fragments, except CNV, as a single score again? Why was a single score allowed to contribute more than others (i.e., logistic score and CNV) to PCM score?
• Line 182: Specify the thresholds.
• The procedures for quantifying fragments, motif, nucleosome footprint
• How are regions for quantifying fragments defined?
6. Was the CNV detection method developed in this study? If so, it would be great to compare it with other CNV detection algorithms. I believe the authors can use the data (referred to as other types of cancer in the manuscript) in this study to do this.
7. How many features are discarded through LASSO regression? The terms “marker” and “feature” are not consistently used throughout the manuscript. Additionally, it might be misleading to refer to this process as “dimensionality reduction”.
8. What is the reason for not performing multiple testing corrections?
9. Providing the processed nucleosome footprint, fragments, and motifs for each sample as the supplementary table would be helpful to ensure the reproducibility of the research without disclosing sensitive information.
10. I recommend caution with the term “predictive”, which often refers to a biomarker that predicts response to treatment, e.g., in the statement: “CfDNA shows predictive prognostic value for pancreatic cancer”. They are really describing a prognostic biomarker.
11. The data are not publicly available “due to patient privacy requirements but are available upon reasonable request from the corresponding author.” It is almost always possible to release the genomic locations of mapped fragments at least, which contain no genetic information. I would strongly encourage this.

Minor comments
Line 101: What is the reference for relationship between early genesis of cancer and ctDNA?
Line 117: Is 227 the correct number of patients?
Figure 1a: What is the meaning of high Z scores in pancreatic cancer?
Figure S1: How is the fragment length profile converted into a single data point? Is it the median?
Table 1 and 2: It would be better to describe table 1 and 2 more clearly.
Line 273: Information on other types of cancer are missed (e.g., how they are collected and processed, how many samples).
Figure S6: Why is maximum scale of y-axis (PCM score) 1?
Figure S7A: Isn't the combo model in Figure 3 the same as the PCM in Figure S7A? Why are the AUCs different?
PCM score in supplementary table 10 is wrong. It must be a multiple of 0.5. PCM score for healthy samples are missing.
The authors use serum and plasma interchangeably it seems. These are very different in the liquid biopsy field, as some

assays work on one but not the other. Please clarify.
There are grammatical errors that could benefit from close copy editing.

Reviewer #3

(Remarks to the Author)
I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Reviewer #4

[Editorial Note: Please also see the attachment displayed on the final three pages of this file]

(Remarks to the Author)
Wu and colleagues introduce a new approach for the earlier detection and prognosis of pancreatic cancer. The approach is based on a large number of samples (n=975) and uses clearly delineated training, validation and external cohorts consisting of pancreatic cancer, patients with benign tumors, chronic pancreatitis and healthy controls. In addition to using cell free DNA for their approach the authors assess other common biomarkers (CA19-9 and CEA). I greatly enjoyed reading the manuscript and think it is an advance that should be published. I think some edits would greatly improve the readability and clarify the results presented.
Major:
• In Figure 1A the samples are illustrated on the chromosome arm level but Figure 1E the rankings are based on chromosomes. Could you split the ranking by arm instead of chromosome and could you indicate whether the copy number alteration was a gain or a loss. Arm level events are more relevant in cancer than the entire chromosome. Also, some arms the events are opposing directions (ie 8p loss and 8q gain).
• According to The Cancer Genome Atlas (taken from fire browse and image reproduced below) the most common aneuploidies in pancreatic cancer are: 1q gain (28%), chr6p loss (36%), chr6q (46%), 7p gain (29%), 7q gain (27%), 8q gain (25%), 9p loss (42%), 9q loss (25%), 20q gain (25%). Could you explain why your 3 top features (chr5, chr4, and chr15) are not typically observed in pancreatic cancer? Also, could you explain why chr6, chr20 and chr7 are very common in pancreatic cancer (as assessed by the TCGA) but at the bottom of your rankings?
• Please add the copy number score to Figure 3 and Supp Table 10.
• “According to the distribution of the Zscore of healthy people in the region defined that the Zscore greater than 2 or less than -2 was the baseline threshold” (lines 153-154). Please clarify in the main text which cohort of healthy samples were used at this step. Isn’t this a circular analysis that guarantees the specificity is 100%? This would also overestimate specificity if you were setting the copy number thresholds based on healthy samples prior to inclusion in the model.

Minor:
• I don’t really understand this sentence (lines 106-108): “Here, we performed a multi-center, large scale cohort study and employed a state-of-the art- next generation sequencing (NGS) technology to acquire plasma cfDNA end motif, nucleosome footprint, and fragments profiles on cfDNA from all enrolled cases.” Why isn’t copy number listed here. It is clearly a major part of the study and seems to separate out the cases and controls.
• “There were a lot of studies working on the early detection of pancreatic cancer, for example, the extracellular vesicle long RNA, and exosomsomal micro RNAs” Seems like an incomplete list. There are other manuscripts that have used copy number, mutations, methylation, and fragmentomics to identify pancreatic cancer in liquid biopsies.
• Why are there missing scores in the PCP.Score (Column 18) in the Supp Table 10? Also, why are some of the missing scores “NA” but others “#N/A”?
• I think on line 316 it should be prospective not perspective.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
The revision is very shoddy and perfunctory. In addition, as this work is soly based on previous biomarkers developed by others, the novelty could only be justified by a high-performance model. However, both the feature selection and building of the model are not convincing, with many technical concerns.

1. The github page the authors provided does not exist with a 404 error.
2. For data sharing, why not upload the aligned BED format file which does not contain any patient-sensitvie data? (echo Reviewer 2)
3. You said you used "featureCounts" v2.19.1. This is funny, because the latest version of this tool is only v2.0.8. I do not understand how did you get the v2.19.1 version (from the future)?
4. in "Filter Genes and Model Construction", did you do this step before model building? If so, there should be a disclosure of

information to the testing dataset, which is inappropriate.
5. "we considered the direction of CNVs relative to oncogenes and tumor suppressor genes to minimize false positives". Do you mean oncogenes should be amplified and suppressors should be LOH? This is not true, and definition of oncogenes/suppressors are not always correct. For example, in HCC, 1p is commonly LOH and 1q is commonly copy number gain, but samples with opposite trends are also found (Jiang et al. PNAS 2015).
6. You use a panel of healthy subjects to model the copy number for others. In this case, you cannot use them anywhere during downstream model building. In Fig. 1e, the CNV signal is empty for all healthy subjects, I guess this should be due to your inappropriate re-analyzing of these cases. In fact, the signal cannot be so clean.
7. "Nucleosome footprint model was built by bedtools (https://bedtools.readthedocs.io/en/latest/) to calculate the reads distribution and remove uncovered genes at the TSS +/- 2.5kb region." This sentence appears twice in your response letter. Similarly, the WGS data processing part appears twice in your revised manuscript.
8. In "procedure of GC correction", I have never seen such method based on your formula, most people use LASSO. Or, why donot use ichorCNA's normalization?
9. "The formula for the single score indicates that at least two out of three dimensions must exceed their respective thresholds to reach the positive threshold of the final PCM score. This approach is intended to prevent a low score in one dimension from negatively impacting the overall logistic score." Is it just bagging? What is the advantage over GBDT?
10. In "Feature selection", you just removed chrY, which means you kept chrX? If so, how do you handle the dosage differences in male and female samples on chrX?
11. "Procedure for quantifying end motif (based on methods in 2018 PNAS, and 2020 Cancer Discovery [9, 10])", there is NO 2018 PNAS paper, and your citation 9 is also incorrect. There are also other incorrect citations in the manuscript, e.g., citations 9 and 31. This made me doubt the familiarity to the field of the authors.
12. You show that your CNV score is correlated with ichorCNA. However, you can see that the regression line is biased by the samples with high tumor DNA fraction. You should use log-scaled values, or focus on those with 5% or less ichorCNA scores, as these are more important early-stage samples.
13. For the disconcordence between your CNA analysis and TCGA, your explaniation is that you are working on cfDNA while TCGA is working on tissue. This is definitely unacceptable, as CNAs inferred from plasma is known to be highly consistent with tumor tissues. Tumor heterogenety and patients' races could be more reasonable explainations.
14. Typos, e.g., "Tabel S3-S5" in line 423.

(Remarks on code availability)
The github page is not available, and I cannot view the code.

Reviewer #2

(Remarks to the Author)
The authors did a good job of addressing my concerns. I am satisfied.

(Remarks on code availability)

Reviewer #3

(Remarks to the Author)
I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

(Remarks on code availability)

Reviewer #4

(Remarks to the Author)
Response: Thank you for your suggestions. We have reviewed additional literature and compared it with our research, we have discussed these comparisons in the Discussion section
• The provided list seems to ignore much of the published literature for the detection of pancreatic cancer in plasma.
Fragmentomes: There was few research using fragmentomes to detect pancreatic cancer, but there was research focusing on liver cancer. Zachariah H. Foda, et.al used cfDNA fragmentome features, achieved high specificity in detecting liver cancer [12]
• That is not the correct citation. This is the correct citation (Genome-wide cell-free DNA fragmentation in patients with cancer –Cristiano et al 2019). The Velculescu lab (DELFI group) looked at fragmentation patterns in pancreatic (n=34) cancers in their 2019 manuscript.
Copy number
CNV is generally not used alone as a diagnostic feature for cancer due to its low sensitivity. But there was research focusing on CNV-based prognostic model. For example, Qian Zhan et al. using CNV to classify pancreatic cancer patients, and a prognostic model was established [13].
• The Vogelstein group has done extensive work on aneuploidy/CNAs as a biomarker in pancreatic cancer (and other cancers). (Assessing aneuploidy with repetitive element sequencing Douville et al 2020). That manuscript evaluates plasma from patients with pancreatic cancer (n=82) and reports sensitivity for copy number analysis. They also did a follow-

up study looking at copy number alterations and integrating it with alu sines and proteins in 287 plasma samples from patients with pancreatic cancer (Machine learning to detect the SINEs of cancer--Douville et al 2024).

• "CNV is generally not used alone as a diagnostic feature for cancer due to its low sensitivity."—This statement is irrelevant to whether copy number alterations have been evaluated in pancreatic cancer or not. It is important to note that many liquid biopsies papers have suggested integrating multiple analytes due to enhance sensitivity. In fact, your manuscript is specially making the argument that you need to integrate end motif, nucleosome footprint, fragmentation, and copy number to garner sufficient sensitivity to detect pancreatic cancer.

Mutations Zill et al. conducted a prospective analysis of five genes (KRAS, TP53, APC, FBXW7, and SMAD4) in tumor tissues and ctDNA from 26 pancreatic cancer patients. Among the 17 patients with successful sequencing, 90.3% (95% CI 73.1%–97.5%) of the mutations detected in tumor tissue were also identified in ctDNA. The diagnostic accuracy of ctDNA sequencing was 97.7%, with an average sensitivity of 92.3% and a specificity of 100% for the five genes [14].

• Cohen et al 2017 (Combined circulating tumor DNA and protein biomarker-based liquid biopsy for the earlier detection of pancreatic cancers) should be included because it is one of the seminal works on pancreas mutations in plasma. This manuscript evaluates mutations in plasma from 221 patients with pancreatic cancer.

Methylation:

• Grail's paper should be cited (Sensitive and specific multi-cancer detection and localization using methylation signatures in cell-free DNA—Liu et al 2020). They have done extensive analysis on detection (and also prognostication) of pancreatic cancers using methylation.

Minor:

• Copy Number Variation (CNV) refers to germline events. Copy Number Alteration (CNA) refers to somatic events. The terminology should be consistent with the field.

(Remarks on code availability)

Version 2:

Reviewer comments:

Reviewer #1

(Remarks to the Author)
This version has improved a lot, but there are still many grammar errors in the response letter and main text that you need to fix.

For the descriptions on Cristiano and Cohen stduies, I suggest move them to Introduction rather than put in Discussion.

The Cristiano dataset has been reanalyzed in various studies, inclduing An et al. Nature Communications 2023 and Ju et al. Cell Reports Methods 2024 with improved performance on pancreatic cancer. In addition, Bie et al. Nature Communications 2023 also investigated pancreatic cancer with high accuracy. You should discuss/cite these papers.

You should also provide the name of your ethics committee.

(Remarks on code availability)
May add more documentation.

Reviewer #4

(Remarks to the Author)
Authors have addressed my concerns.

(Remarks on code availability)

Response to Reviewer #1:

Reviewer #1 (Remarks to the Author):

Wu et al. reported cfDNA-based models for pancreatic cancer diagnosis and prognosis prediction. The topic is very important, however, the manuscript is terribly written, very difficult to follow, and many details are hidden. Since both the data and codes are unavailable, it is impossible to evaluate the reliability of the results.

Response: We gratefully appreciate for your valuable comments. In the revised version, we have carefully edited the manuscript and we sincerely hope that it will improve the quality of our manuscript. Additionally, we have provided as much detail as possible regarding the content of our research, and we have included both our raw data and codes. The raw data has been deposited to the CNSA database (https://db.cngb.org/cnsa/), project number: CNP0005903. The codes are available on GitHub (https://github.com/JimmyWu2024/PAAD_pipeline). All raw data and code access details are provided in the manuscript under the **Data and Code Availability** section.

Detailed comments:

1. *There is no need to have a Suppl. Method, the authors should move the related content to main text.*

Response: Thank you for your kind suggestion. We have moved the content from the Supplementary Methods to the main text, under the Materials and Methods section. Additionally, to ensure the reproducibility of our results and in accordance with the suggestions from you and the other reviewers, we have added more detailed descriptions in the Materials and Methods section. These include the procedures for fragment quantification, end motif analysis, nucleosome footprinting, and CNV score calculation (line 325-line 384).

2. for NF, the methods part "Nucleosome footprint model was built by bedtools to calculate the reads distribution and remove uncovered genes at the TSS +/- 2.5kb region". This is far from complete, and I do not understand how to calculate NF. Is it a value, or a vector of many values?

Response: We apologize for not clearly explaining the NF calculation process in the manuscript. In fact, our method is based on research based on *Cell* (2016) [1] and *Nature Genetics* (2016) [2]. Based on these studies, we developed our own NF calculation process, which consists of five steps:

(1) **Acquisition of the Promoter Regions of the Whole Genome:** Promoter regions were identified using transcription start sites (TSS) of the main transcripts of reference genes published in the UCSC database are used (https://genome.ucsc.edu/), with 2500 bp extended upstream and downstream as the promoter region of the gene;

(2) **Defining the Central and Peripheral Regions:** The central region of the promoter is defined as the 250 bp immediately adjacent to the transcription start site of the

gene, while the peripheral region was extended 2500 bp on either side of the central region. This classification is based on the observation that actively transcribed genes tend to have sparser nucleosome distribution near the TSS, making them more susceptible to degradation once in the bloodstream. As a result, sequencing depth is expected to be lower in the central region compared to the peripheral region.

(3) **Obtaining Region Coverage and Sequencing Depth:** The software "Bedtools" (v1.6.2) was used to calculate region coverage, while "featureCounts" (v2.19.1) was employed to determine sequencing depth for each region, which was then converted to FPKM.

(4) **Quantifying the Differences in Nucleosome Distribution for Each Gene:** The nucleosome distribution difference score is calculated as the sequencing depth of the peripheral region (FPKM) minus the central region (FPKM). This score represents the distribution of nucleosomes and the transcriptional activity of the gene.

(5) **Filter Genes and Model Construction:** After removing housekeeping genes and silenced genes from the whole genome, 20315 genes remained. Genes covered in at least 90% of the samples were retained, and the rank-sum test was applied to calculate p-values. Genes with p-values ≤ 0.01 were further filtered, resulting in 428 genes. LASSO was then used for dimensionality reduction, ultimately selecting 102 genes for model construction.

The content mentioned above has been added to the main text of the manuscript (line 342-line 364), and we hope it addresses your concerns. Should you have any further questions, we would be happy to provide additional explanations.

3. The authors used 20kb bins to screen for CNVs, is it too small that may cause high variations?

Response: Your concern is very valid, and we have taken it into careful consideration. We apologize for not detailing the CNV screening process in our manuscript. To clarify, the 20 kb length is the basic unit used for CNV calculation, not the final reported CNV length. To reduce variations associated with smaller bins, we merged adjacent bins that met specific criteria, allowing for a margin of error during this process. The final reported CNV length is at least 2 Mb, and any length below 2 Mb is filtered out. In addition, we calculate CNV score based on CNV region, following the method reported in a 2013 *Cell* paper [3]. During this process, we considered the direction of CNVs relative to oncogenes and tumor suppressor genes to minimize false positives. We selected smaller bins to ensure high specificity and achieve precise boundaries for the CNV regions.

4. To calculate CNVs, a group of healthy subjects are used. How many? Are these samples also used in diagnosis, either training or testing?

Response: The healthy samples we calculate CNVs were healthy samples in training cohort, the number of these healthy samples were 160. These healthy samples were also used for training the diagnosis model.

5. "To obtain the best surveillance model". I am confused, as the logistic regression is designed for diagnosis?

Response: We apologize for the error in our wording. It should have been "diagnostic model." We have made the correction in the manuscript. (line 423)

6. Their are lots of formula, how did not determine the parameters? e.g., 2.13, 3.26, 2.85 in Logistic Score?

Response: The feature weights of logistic score and PCP score in the training cohort is the weight determination of logistic regression, which involves the following key steps:
(1) **Establishing a Linear Combination:** A linear combination is formed between the categorical labels and the probability values generated for each dimension of each sample.
(2) **Probability Conversion:** The sigmoid function is applied to convert the linear combinations into probability values.
(3) **Weight Determination:** Maximum likelihood estimation is used to determine the weights for each dimension.
(4) **Gradient Descent:** Feature weights are updated using the gradient descent method.
(5) **Iterative Optimization:** This process is repeated iteratively until convergence is achieved.

7. For SVM, is it just used for feature selection? And you used its model to get calculate an intermediate feature, then artefically mixed it with the features used in SVM again to build a PCM score. This is very confusing and makes no sense. Why not build a diagnostic model using the features directly?

Response: We apologize for not clearly explaining the use of SVM in the article. SVM was employed for model training, while LASSO was utilized for feature selection. Initially, SVM was applied separately to each of the three dimensions (NF, Fragment, Motif). The models created with the selected features for each dimension are independent and each produces a result. Subsequently, logistic regression is used to combine the output results from these three dimensions. The final PCM score incorporates the results from all three dimensions alongside CNV, with its composition detailed in Table S3. We chose not to directly merge features from multiple dimensions at the outset due to the distinct genomic information each feature represents, including significant variations in absolute value ranges and biological significance. Thus, we opted to build separate models using the same machine learning algorithm for each dimension before integrating their outputs. This methodology is based on research published in the 2014 *New England Journal of Medicine* [4] and has been rigorously validated with large datasets in our prior studies [5, 6], yielding excellent results. Consequently, we decided to adopt this approach in our current research.

8. There are duplicated sentences in "Patients and cohorts".

Response: Thank you for pointing out the duplicated sentences in the '**Patients and**

Cohorts' and '**Materials and Methods**' sections. We apologize for the oversight and have removed the redundant content. Additionally, we believe it is clearer to combine the '**Patients and Cohorts**' content with the relevant details in the '**Materials and Methods**' section to enhance coherence.

9. "We also collected serum from patients with other types of cancer". This is funny, as your model is built using plasma, but you validate it using serum?

Response: We apologize for the error in our wording. It should have been "plasma" instead of "serum". We have made the correction in the revised version of our manuscript.

10. Many sentences in Discussion should be put into Introduction, as they are background information, and the authors did not discuss the results using them.

Response: Thanks for your suggestion, we have moved the background information in **Discussion** section to **Introduction** section. In addition, to make the **Discussion** section more comprehensive, we have also added the following content.

(1) As you mentioned in question 11, we added the advance of our work over the previous work; including ctDNA-based liquid biopsy and circular tumor cell (CTC)-based liquid biopsy;

(2) We added the advantages of plasma cfDNA in liquid biopsy: the detection technology is mature, and cfDNA is relative stable.

11. What is the advance over the previous works on pancreatic cancer (ref. 26-28). This part actually should be discussed in detail.

Response: Thanks for your suggestion. Compared to previous studies [1-3], we believe our study offers a more comprehensive analysis of cfDNA characteristics, improving the identification of cfDNA features associated with pancreatic cancer progression. Most previous studies have focused on single cfDNA feature types. For example, Ying L et.al used a methylation-based cfDNA signature, while Guler GD focused on a 5-hydroxymethylcytosine signature. Compared to previous studies, our study introduces several key advancements:

1. We incorporated four types of cfDNA features, including the fragment length, nuclear footprint, end motif and CNV.
2. Our study enrolled substantial number of patients with benign pancreatic tumors, enabling our model to distinguish not only between pancreatic cancer and non-pancreatic cancer but also between pancreatic cancer and benign pancreatic tumors.
3. We developed a prognostic model based on cfDNA signatures, addressing the gap in previous studies that have primarily focused on diagnostic rather than prognostic models.

We have added this discussion to the manuscript, and the changes are highlighted in red.

12. "next generation sequencing (NGS) technology to acquire plasma cfDNA end motif

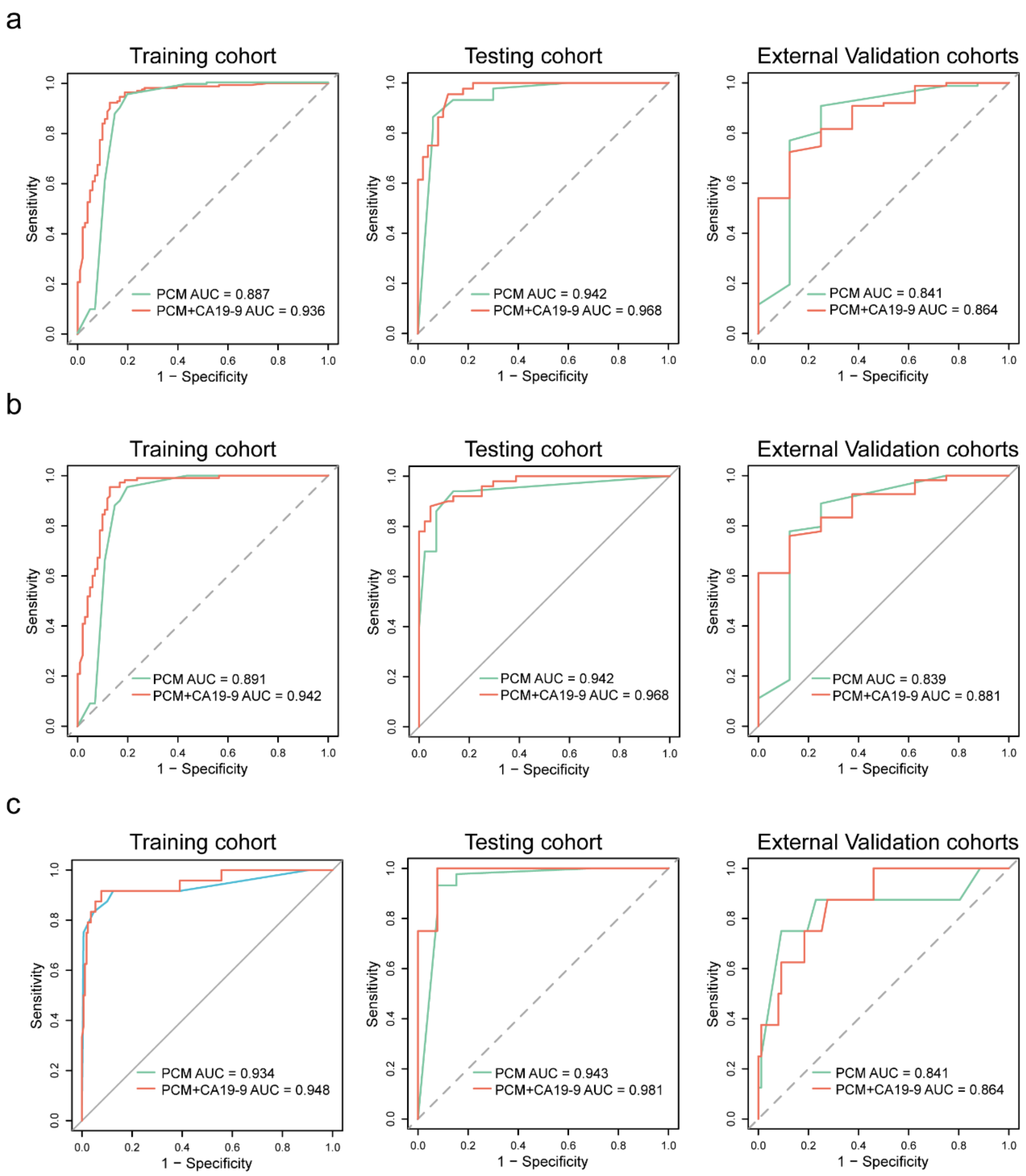
[29]". Citation 29 is "Branco MR, Ficz G, Reik W. Uncovering the role of 5-hydroxymethylcytosine in the epigenome. Nat Rev Genet 2011; 13: 7-13", why cite this one?

Response: Thank you for bringing this to our attention. We apologize for the incorrect citation. The correct reference is "Jiang, P. et al. Plasma DNA end motif profiling as a fragmentomic marker in cancer, pregnancy and transplantation, Cancer Discov 10, 664-673", we have updated this in our manuscript. (line 98)

13. Would the performance improve by integrating the protein biomarkers?

Response: Yes, as CA19-9 is a well-established biomarker for detecting pancreatic cancer, we included it in our model in the original manuscript. The performance of the PCM score combined with CA19-9 is shown in Fig. S7, with the equation provided in the main text (lines 259-261). As illustrated in Fig. S7, integrating CA19-9 enhanced the model's performance across the training, testing, and validation cohorts.

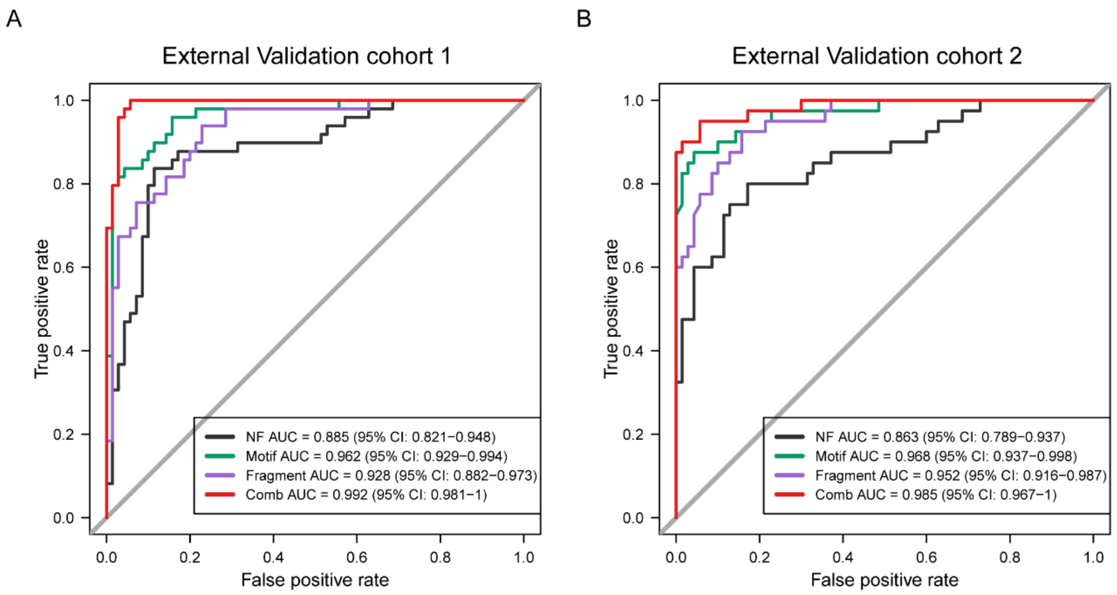
Fig. S7



Reviewer #2 (Remarks to the Author):

This study constructed models for early detection and prognosis of pancreatic cancer using features of cell-free DNA (cfDNA) in plasma and validated these models by using two external cohorts. Early detection of pancreatic cancer is a clinically important unmet need and this study is timeline. However, insufficient documentation and supporting results significantly hinder transparency and reproducibility of the research. Another concern is that the so-called external cohorts are not genuinely external since the healthy samples assigned as external cohort were processed and collected in the same way as training cohort. Therefore, this might overestimate generalizability of their model, as it is not truly unseen data. Furthermore, the documentation of this study requires substantial improvement. Most of computational procedures are mostly unmentioned in the methods section. Additionally, some computational code/file they distributed is encrypted and cannot be opened.

Response: Thank you for your valuable feedback. In the revised manuscript, we have provided additional documentation and supporting results. The healthy samples in our study were collected from multiple institutions. To address concerns regarding the model's generalizability, we collected an additional 65 healthy samples from a different institution. We calculated the positive rate of these samples and re-evaluated the model's performance in the external validation cohorts using this new data. The PCM scores indicated that none of these samples were classified as positive. We also replaced the healthy samples in external validation cohorts 1 and 2, and the model's performance in both cohorts remained robust.
The raw sequencing data of these healthy samples we have uploaded in CNSA database (https://db.cngb.org/cnsa/), project number: CNP0006304.
Additionally, we refined the **Methods and Materials** section to provide the most detailed methodological information possible. We also re-uploaded the code to ensure that all code is publicly accessible. For the re-uploaded code, you can download in Github (https://github.com/JimmyWu2024/PAAD_pipline).



Performance evaluation of NF, motif, fragment features, and combined model (PCM score) in classification of pancreatic cancer and non-cancer groups with new healthy samples.
a.ROC curve analysis for the NF, motif, fragment and combined model (PCM score) in External Validation cohort 1. **b**. External Validation cohort 2.

Major comment
1. Limit of detection of the model
It is well known that tumor content in cfDNA significantly affects the detection limit of cancer and any other cfDNA applications. Previous research using cfDNA have already demonstrated that cancer detection in samples with a high amount or proportion of circulating tumor DNA (ctDNA) is relatively easy. I recommend the authors testimate

ctDNA content using other tools (e.g., ichorCNA [PMID: 29109393]) and report an approximate limit of detection in terms of ctDNA content.

Response: Thanks for your valuable suggestion. Yes, our research findings may be influenced by the ctDNA content. Based on your suggestion, we utilized the ichorCNA tool to testimate ctDNA content. According to the results of ctDNA, and the results revealed the following distribution among our samples: 56.82% had a ctDNA/cfDNA content below 1%, 27.60% ranged between 1% and 1.5%, 9.74% between 1.5% and 2%, and only 5.84% exceeded 2%. For patients with pancreatic cancer, the average ctDNA/cfDNA ratio was 1.7%, whereas, in patients with PBT, the ratio was 0.9%. Overall, the ctDNA/cfDNA ratio is relatively low, with a detection limit of 0.5% for ctDNA content. The ctDNA/cfDNA was added to Table S10. We believe its impact on the cfDNA diagnostic model is limited.

2. Among the main results, only figures 4b,c support the possibility of early detection by the model. It would be great to evaluate each cohort separately (testing, external 1 and 2) in the analysis in figures 4b,c.

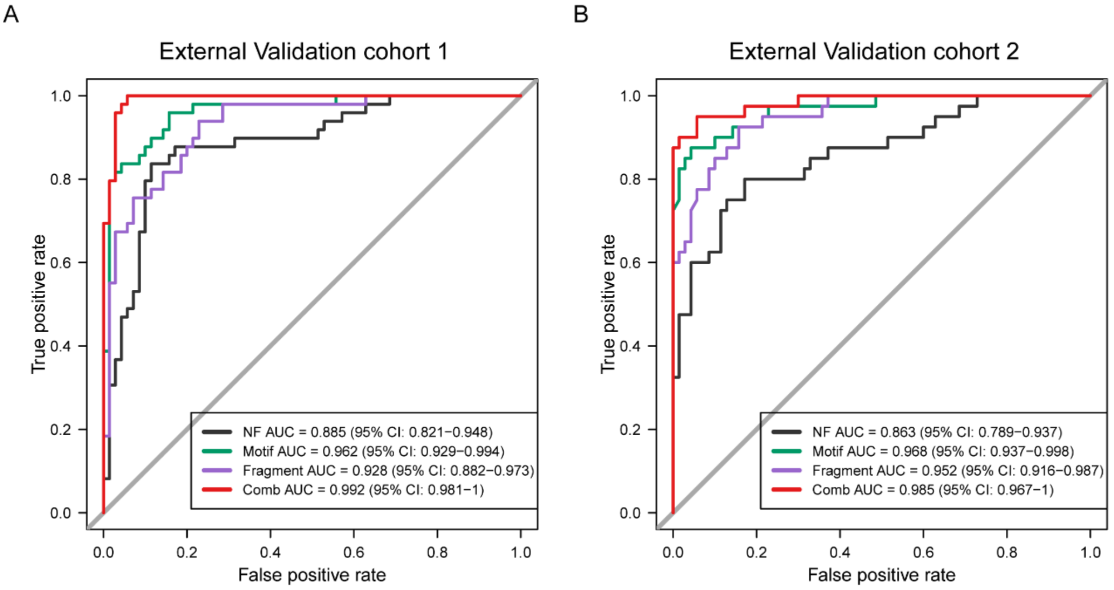
Response: Thank you for your suggestion. In fact, in our original version of manuscript, we have already separated the cohorts in Figure 4b,c and calculated the AUC, specificity, sensitivity, and accuracy for each cohort in Table 2.

3. Rather than simply providing a KM plot, it would be helpful to show how many high-risk patients can be accurately identified in the external cohort using the PCP score and defined threshold (i.e., 0.3127).

Response: Thanks for your valuable suggestion. If we defined patients who experienced death or recurrence within one year as high-risk patients, then in External Validation Cohort 1 (40 patients), there were 19 patients in high risk, of whom 17 were correctly identified based on our threshold, resulting in an identification accuracy of 89.5%. In External Validation Cohort 2 (31 patients), there were 18 high-risk patients, and all patients were correctly identified using our threshold, yielding an accuracy of 100%. We have included this content to the manuscript. (line 210-line 215)

4. The healthy samples in training and external validation cohort are collected in the same way. It seems they just randomly distributed to training/testing/external validation cohorts. This is a problematic design that could allow leakage of information into the external validation cohorts. For instance, if all of the healthy samples vary is some way that is NOT generalizable and not due to the absence of cancer, the model may seem to perform well on external validation, but fail when applied to truly “unseen” cohorts.

Response: We fully understood your concern. In fact, the healthy samples were collected from different institutions. To further address your concerns, we recollected 65 healthy samples from an additional institution. After calculating the PCM scores for these samples, we found that none were classified as positive. We also replaced the healthy samples in external validation cohorts 1 and 2, and the model’s performance in both cohorts remained robust. The ROC curves and detailed information of the recollected samples are shown below. The raw sequencing data of these healthy samples we have uploaded in CNSA database (https://db.cngb.org/cnsa/), project number: CNP0006304.



Performance evaluation of NF, motif, fragment features, and combined model (PCM score) in classification of pancreatic cancer and non-cancer groups with new healthy samples.

a.ROC curve analysis for the NF, motif, fragment and combined model (PCM score) in External Validation cohort 1. **b**. External Validation cohort 2.

Table 1 Detailed information of the recollected healthy samples

Sample	Age	Gender	PCM score	NF score	Fragment score	motif score	LR score	CNV score	CNV num	IchorCNA(ctDNA ratio)
Healthy499	47	M	0	0.06295808	0.5	0.21433721	0.08957273	0	0	0.004991
Healthy500	56	F	0	0.32039142	0.15354676	0.09312853	0.04094975	0	0	0.006882
Healthy501	57	F	0	0.00931092	0.00687119	0.07359744	0.01342931	0	0	0.008415
Healthy502	52	F	0	0.37036116	0.09455499	0.31644129	0.07671221	0	0	0.005768
Healthy503	50	M	0	0.34307155	0.13151993	0.06259268	0.03669879	0	0	0.006107
Healthy504	65	M	0	0.1829957	0.19192044	0.12469211	0.03792053	0	0	0.008468
Healthy505	50	F	0	4.00E-06	0.0066041	0.04865179	0.01214494	0	0	0.007683
Healthy506	61	F	0	0.35121758	1.96E-07	0.14636864	0.03382466	0	0	0.004092
Healthy507	51	F	0	0.07235259	0.43637813	0.25115667	0.08625584	0	0	0.004213
Healthy508	66	M	0	0.4637656	0.4898999	0.13862293	0.14906775	0	0	0.00689
Healthy509	48	M	0	0.17560688	0.08342524	0.16439153	0.03139637	0	1	0.008777
Healthy510	47	F	0	0.21989201	0.09413929	0.0793917	0.02708328	0	0	0.005953
Healthy511	62	F	0	0.14674845	0.06676203	0.16143657	0.02798643	0	0	0.006983
Healthy512	46	M	0	0.04298832	0.10537523	0.07300212	0.01895781	0	0	0.008616
Healthy513	69	M	0	0.36123751	0.36172722	0.05200929	0.06866171	0	0	0.008321
Healthy514	61	M	0	0.1595562	0.04377336	0.04584873	0.01866043	0	0	0.009356
Healthy515	67	F	0	0.02852653	0.1017211	0.03749469	0.01625041	0	0	0.009442
Healthy516	61	F	0	0.04287551	0.02920375	0.10258285	0.01683469	0	0	0.007597
Healthy517	55	F	0	0.04875306	0.31336779	0.08764773	0.03578824	0	0	0.009235
Healthy518	50	M	0	0.45707337	0.11681725	0.04536779	0.04216059	0	0	0.004808
Healthy519	47	M	0.5	0.87623916	0.42985427	0.161077	0.27634039	0	0	0.008546
Healthy520	45	F	0	0.03575868	0.07622958	0.04308072	0.01563757	0	0	0.00795
Healthy521	46	F	0	0.16231121	0.07917827	0.33056052	0.05077906	0	0	0.008958
Healthy522	60	F	0	0.13057408	0.21325211	0.30958041	0.06404831	0	0	0.008841
Healthy523	49	F	0	0.13297189	3.58E-06	0.08657169	0.01779107	0	0	0.009672
Healthy524	57	M	0	0.06798412	0.03458334	0.0887306	0.01722975	0	0	0.006665
Healthy525	51	F	0	0.00386112	0.01146175	0.14318076	0.0168138	0	1	0.009443
Healthy526	41	M	0	0.0042774	4.40E-06	0.10620835	0.014474	0	2	0.009886
Healthy527	47	M	0	0.0291594	0.07893799	0.09519352	0.01836363	0	0	0.008483
Healthy528	45	M	0	0.15141018	0.01062362	0.11385487	0.0207805	0	0	0.008409
Healthy529	65	M	0.5	0.77930321	0.00484391	0.3751519	0.15675658	0	0	0.004353
Healthy530	53	F	0	0.03112676	0.07806141	0.26639901	0.03169557	0	0	0.005513
Healthy531	66	F	0	0.32009998	0.03650862	0.08915655	0.02929258	0	0	0.003713
Healthy532	50	F	0	0.11718704	0.21550472	0.07755169	0.03047397	0	0	0.004099
Healthy533	53	M	0.5	0.89044184	0.1283743	0.16636028	0.14499349	0	0	0.007982
Healthy534	51	F	0	0.03232497	0.26509039	0.10769612	0.03226765	0	0	0.006273
Healthy535	51	M	0	0.02143207	0.27673492	0.03245725	0.02568061	0	0	0.006935
Healthy536	67	F	0	0.01309073	0.03862711	0.04296892	0.01341271	0	1	0.006769
Healthy537	65	M	0	0.03154819	0.07124739	0.07393624	0.01687435	0	1	0.00852
Healthy538	41	F	0	0.09357294	0.04051018	0.11934705	0.02037864	0	0	0.007677
Healthy539	49	F	0	0.03585621	0.15921749	0.17180831	0.02970702	0	0	0.003817
Healthy540	55	M	0	0.39248468	0.48756288	0.08431132	0.11133929	0	0	0.006262
Healthy541	70	F	0.5	0.78558638	0.01977849	0.00843414	0.0561777	0	0	0.007732
Healthy542	55	F	0.5	0.05606944	0.61420213	0.0424813	0.07122814	0	0	0.007683
Healthy543	68	M	0	0.47895148	0.09079391	0.15461811	0.05760117	0	0	0.005879

Healthy544	59	F	0.5	0.96093153	0.21776527	0.21254576	0.22806102	0	0	0.009645
Healthy545	67	M	0	0.32917116	0.0679433	0.04605886	0.02840894	0	0	0.007046
Healthy546	56	F	0	0.01625318	0.3864242	0.07971158	0.0399073	0	0	0.007956
Healthy547	52	M	0	0.34652393	0.12727127	0.05326561	0.03547608	0	0	0.004627
Healthy548	64	F	0.5	0.07327484	0.58587207	0.05386779	0.07076341	0	0	0.006926
Healthy549	48	F	0	0.16472423	0.16559752	0.07959578	0.02947193	0	0	0
Healthy550	50	F	0	0.04422737	0.00542914	0.15931619	0.01893969	0	0	0.007743
Healthy551	57	F	0.5	0.75170727	0.02054341	0.0584996	0.06133122	0	0	0.006776
Healthy552	43	M	0	0.21250866	1.10E-06	0.18960539	0.02913416	0	0	0.009586
Healthy553	61	F	0	0.0255859	0.09541641	0.12422988	0.02094091	0	0	0.007922
Healthy554	63	M	0	0.00564417	0.07962285	0.11651838	0.01875375	0	0	0.009624
Healthy555	61	F	0	0.07836563	0.04804969	0.05999184	0.01667686	0	0	0.008552
Healthy556	46	F	0	0.28018078	0.26868544	0.27875993	0.09061074	0	0	0.005715
Healthy557	51	M	0	0.00557359	5.52E-06	0.09512794	0.01400603	0	0	0.005307
Healthy558	51	F	0	0.02726266	0.05800702	0.05319833	0.01507759	0	0	0.004558
Healthy559	41	M	0	0.09670471	0.16965347	0.02413682	0.02170672	0	0	0.007998
Healthy560	54	F	0	0.05194147	0.40457908	0.02238169	0.03770175	0	0	0.007218
Healthy561	45	M	0.5	0.13144727	0.600646	0.02809728	0.07633871	0	0	0.008629
Healthy562	46	M	0	0.2204792	0.01607782	0.13997022	0.02646208	0	0	0.008931
Healthy563	58	M	0	0.02751025	0.00515468	0.02710753	0.01195653	0	0	0.005999

5. The method section could use more detail. While certain analysis and data are mentioned in the results, they are completely omitted in the method section and supplementary methods. Particularly, Python, R and Perl code can be found in their GitHub repository, but only Python programs are mentioned in the manuscript. At the very least, the following items must be clearly explained:

Response: Thank you for bring this to our attention. We have reorganized our codes and added more detailed information based on your feedback and the recommendations from other reviewers. The codes are accessible on GitHub (https://github.com/JimmyWu2024/PAAD_pipeline). Additionally, we realized that we indeed missed mentioning some software in the **Statistical Analysis** section. We have now included these in the revised manuscript, which are as follows:

"Fastq files were processed by fastp software (https://github.com/OpenGene/fastp).

Clean data were aligned to human reference genome GRCh37 using bwa-aln (https://github.com/lh3/bwa).

Duplicate reads were marked by sambamba (https://github.com/biod/sambamba/).

Samtools (http://samtools.sourceforge.net/) was used to calculate mapping rate, duplicate rate and genome coverage.

The bam files were further filtered by Samtools, removing unmapped reads, low quality reads, marked duplicates and sequences with no perfect match between read1 and 2.

Nucleosome footprint model was built by bedtools (https://bedtools.readthedocs.io/en/latest/) to calculate the reads distribution and remove uncovered genes at the TSS +/- 2.5kb region.

Nucleosome footprint model was built by bedtools (https://bedtools.readthedocs.io/en/latest/) to calculate the reads distribution and remove uncovered genes at the TSS +/- 2.5kb region.
Base mismatch model was built by VarScan (http://varscan.sourceforge.net) to detect single base mutation.
Fragmentation model was built by pysam (https://pysam.readthedocs.io/en/latest/) to calculate the length of insertion fragment and ratio of short/long fragment in different regions.
The following software and packages were used in our study:
Python (v.2.7.16)
python package 'sklearn'(v1.3.1)
R (v.3.6.3)
R package 'pROC' (v1.16.2);
R package 'datatable' (v1.14.2)
We have added this information to the manuscript, and the changes are highlighted in red. (line 463-line 475)

- *line 150: procedure of GC correction*

Response: GC bias can affect the results of CNV analysis, so correction is necessary. The correction process was calculating the average depth of bins for each GC content, then computing the overall average depth of all bins to correct the sequencing depth. The formula is:

$$\text{Depth of corrected bin} = \frac{depth\ of\ all\ bins}{the\ average\ of\ bins\ which\ has\ the\ same\ GC} * depth\ of\ the\ bin$$

- *line 154: The rationale of the Z-score threshold. Is it a threshold which healthy samples do not present CNV?*

Response: Yes. The threshold of Z-score is the value which healthy samples do not present CNV.

- *Line 160: Why are CNV scores for tumor suppressor genes and oncogenes are calculated separately?*

Response: As the reported CNVs are greater than 2 Mbp in length, they may encompass both tumor suppressor genes and oncogenes. For CNVs with an increased copy number, oncogenes should predominantly contribute to align with biological significance, and the reverse applies for tumor suppressor genes. This calculation process follows the methodology outlined in 2013 *Cell* publication [3].

- *Line 176: What is the rationale for nucleosome footprint, motif and fragments, except CNV, as a single score again? Why was a single score allowed to contribute more than others (i.e., logistic score and CNV) to PCM score?*

Response: The formula for the single score indicates that at least two out of three dimensions must exceed their respective thresholds to reach the positive threshold of the final PCM score. This approach is intended to prevent a low score in one dimension from negatively impacting the overall logistic score. The data also reveal instances where some samples show a low logistic score and no CNV, yet two out of the three dimensions still exceed their thresholds. This strategy aims to enhance sensitivity while minimizing the loss of specificity. We have also employed this approach in our previous studies [5].

- *Line 182: Specify the thresholds.*

Response: Thanks for your suggestion. The threshold are as follows: NF score at 0.5051, fragment score at 0.5245, motif score at 0.4677, logistic score at 0.4725, CNV score at 0. We have added this threshold values for each score in Table S10.

- *The procedures for quantifying fragments, motif, nucleosome footprint*

Response: The procedures for quantifying fragment, motif, and nucleosome footprint are outlined below. We have also added this information to the **Material and methods** section. (line 324-line 363)

1. Procedure for quantifying fragments (based on methods in 2018 *Sci Transl Med* and 2019 *Nature* [7, 8]):

- The genome was divided to 3055 regions; the length of every region is 1 Mbp.
- DNA fragments were aligned to the reference genome, and their genomic locations were identified and corrected.
- Fragments between 90 and 150 bp are defined as short fragments, and those between 150 and 220 bp are defined as long fragments.
- The short-to-long fragment ratio was calculated.
- Feature selection: Regions on the Y chromosome and regions with no detected DNA fragment coverage were removed from the initial 3055 regions, leaving 2890 regions. LASSO was then used for feature selection, resulting in 154 regions, which were used to construct the model.

2. Procedure for quantifying end motif (based on methods in 2018 *PNAS*, and 2020 *Cancer Discovery* [9, 10]):

- DNA fragments were aligned to the reference genome to determine start and end positions, followed by correction.
- The 4-mer nucleotide sequence at the 5' end of each fragment was counted.
- Among 256 possible 4-mers, the proportion of each type was calculated.
- Feature selection: LASSO identified 33 motifs to use for model construction.

3. Procedure for quantifying nucleosome footprint (based on methods in 2016 *Cell* and 2016 *Nat Genet*[1, 2]):

- Promoter regions acquisition: Using the UCSC database (https://genome.ucsc.edu/), transcription start sites (TSS) of main reference gene transcripts were identified, extending 2,500 bp upstream and downstream as the promoter region.
- Defining the central and peripheral regions of the promoter: The central promoter region was defined as the 250 bp adjacent to the TSS, with the peripheral region extending 2,500 bp on each side. Actively transcribed genes show a sparser nucleosome distribution near the TSS, where cfDNA degradation is likely, resulting in lower sequencing depth;
- Obtaining region coverage and sequencing depth: “Bedtools” software (v1.6.2)

was used calculate region coverage, and “featureCounts” software (v2.19.1) was utilized to determine the sequencing depth for each region, which was then converted to FPKM;

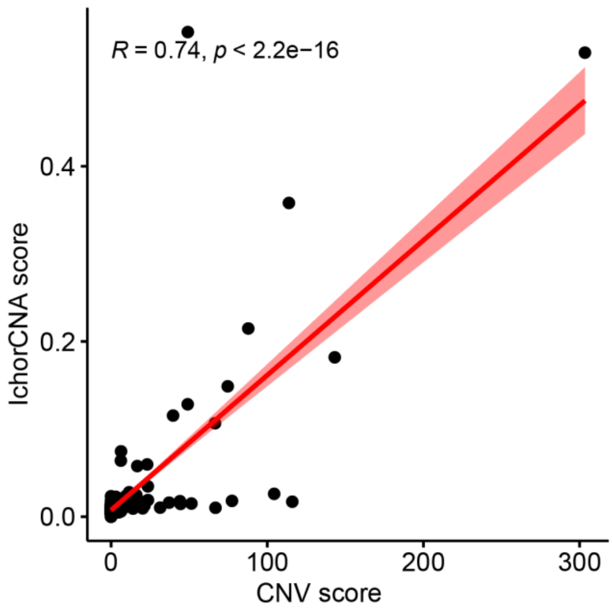
- Quantifying nucleosome distribution differences: The nucleosome distribution difference score is calculated as the sequencing depth of the peripheral region (FPKM) minus the central region (FPKM). This score represents the distribution of nucleosomes and the transcriptional activity of the gene;
- Feature selection: After removing housekeeping and silenced genes, 20,315 genes remained. Genes covered in 90% of samples were filtered by a rank-sum test ($p \leq 0.01$), leaving 428 genes. LASSO further reduced this to 102 genes for model construction.

- *How are regions for quantifying fragments defined?*

Response: The quantifying fragments were defined as those fragments which excluding the Y chromosome regions and centromeric region.

6. Was the CNV detection method developed in this study? If so, it would be great to compare it with other CNV detection algorithms. I believe the authors can use the data (referred to as other types of cancer in the manuscript) in this study to do this.

Response: Yes, the CNV detection method was developed by ourself in our study. Since CNV was not calculated for patients with other types of cancer, we utilized the ichorCNA tool to testimate CNV for every patient in training, testing, and external validation cohorts. We examined the association between the IchorCNA score and our CNV score across samples. The correlation between CNV score and IchorCNA score, shown below, demonstrates good consistency, indicating the feasibility of our method. The IchorCNA score of every patient was added to Table S10.



7. How many features are discarded through LASSO regression? The terms “marker” *and "feature" are not consistently used throughout the manuscript. Additionally, it might be misleading to refer to this process as "dimensionality reduction".*

Response: Thanks for your suggestion, In the process of quantifying end motifs, 223 motifs were discarded by LASSO and 33 motifs remained. For your second suggestion, we have changed the term "marker" to "feature" throughout the manuscript. For your third suggestion, we have replaced "dimensionality reduction" with "feature selection" as recommended.

8. What is the reason for not performing multiple testing corrections?

Response: In our study, all hypothesis tests were based on comparisons between two groups and did not involve comparisons among multiple groups; therefore, multiple testing correction was not required.

9. Providing the processed nucleosome footprint, fragments, and motifs for each sample as the supplementary table would be helpful to ensure the reproducibility of the research without disclosing sensitive information.

Response: We fully agree with your suggestion, in our revised version, we have included the score of nucleosome footprint, fragments, and motifs of each sample in Tabel S10.

10. I recommend caution with the term "predictive", which often refers to a biomarker that predicts response to treatment, e.g., in the statement: "CfDNA shows predictive prognostic value for pancreatic cancer". They are really describing a prognostic biomarker.

Response: Your suggestion is very valuable to us, and we fully agree with your viewpoint. We have made revisions about the "predictive" term in the manuscript.

11. The data are not publicly available "due to patient privacy requirements but are available upon reasonable request from the corresponding author." It is almost always possible to release the genomic locations of mapped fragments at least, which contain no genetic information. I would strongly encourage this.

Response: Thanks for your suggestion. In the revised version of the manuscript, we have provided the original data in the **Data availability** section. The raw sequencing data has been deposited to the CNSA database (https://db.cngb.org/cnsa/), project number: CNP0005903.

Minor comments

Line 101: What is the reference for relationship between early genesis of cancer and ctDNA?

Response: We apologize for omitting this citation, and we have added the reference of the sentence in the manuscript (line 93), the reference is:

Mattox, A. K. et al. The Origin of Highly Elevated Cell-Free DNA in Healthy Individuals and Patients with Pancreatic, Colorectal, Lung, or Ovarian Cancer. Cancer Discov 13, 2166-2179, doi:10.1158/2159-8290.CD-21-1252 (2023).

Line 117: Is 227 the correct number of patients?
Response: Thanks for pointing out our mistakes, actually, there were 370 patients from Changhai Hospital, including 197 PDAC, 17 ASCP, 63 IPMN, 34 PNET, 20 SCN, and 39 CP. We have corrected the number in the revised version of manuscript. (line 129)

Figure 1a: What is the meaning of high Z scores in pancreatic cancer?
Response: The Z score indicates the ratio of short fragments to long fragments, with a high Z score indicating a higher prevalence of short fragments.

Figure S1: How is the fragment length profile converted into a single data point? Is it the median?
Response: The procedure for converting fragment length into a single data point is as follows:
(1) Count the length of all cfDNA fragment in the sample;
(2) Calculate the mean fragment length by dividing the sum of all fragment lengths by the total number of fragments.

Table 1 and 2: It would be better to describe table 1 and 2 more clearly.
Response: Table 1 shows the performance of the PCM score for staged pancreatic cancer versus non-cancer (including PBT, CP and healthy individuals) and pancreatic cancer versus healthy individuals.
Table 2 compares the performance of the PCM score and CA19-9 for staged pancreatic cancer versus pancreatic disease (PBT and CP). As shown, the PCM score outperforms CA19-9 in both the Testing cohort and the two external validation cohorts. Additionally, the PCM scoring system more accurately differentiates early-stage pancreatic cancer compared to CA19-9. We have added further details on Table 1 and Table 2 in the revised manuscript (line 179-line 186).

Line 273: Information on other types of cancer are missed (e.g., how they are collected and processed, how many samples).
Response: Thanks for your suggestion. The samples of colon adenocarcinoma (COAD) were collected from Berry corporation (Beijing, China), with a sample size of 125. Stomach adenocarcinoma (STAD) patients were collected from Berry corporation (Beijing, China), with a sample size of 45. Esophageal cancer (ESCA) patients were collected from Nanfang Hospital, Southern Medical University (Guangzhou, China), and the number of patients were 107. Liver hepatocellular carcinoma (LIHC) patients were collected from the Eastern Hepatobiliary Surgery Hospital, Second Military Medical University (Shanghai, China), with a sample size of 113. The sample processing procedure was consistent with that used for pancreatic cancer samples. We have included this information to the Supplementary appendix.

Figure S6: Why is maximum scale of y-axis (PCM score) 1?
Response: Thank you for bring this to our attention, and we apology for the error, the y-axis represents the logistic score, not the PCM score. We have corrected the title of y-axis.

Figure S7A: Isn't the combo model in Figure 3 the same as the PCM in Figure S7A? Why are the AUCs different?
Response: The comb model in Figure 3 is the same as the PCM score in Figure S7A. The difference in AUCs between the two figures is attributed to variations in the samples. In Figure S7A, the model incorporates CA19-9; however, CA19-9 data were missing for some pancreatic disease patients and all healthy individuals. In Figure 3, the ROC curves distinguish pancreatic cancer from non-cancer cases (including PBT, CP, and healthy individuals), whereas in Figure S7, the ROC curves differentiate pancreatic cancer from other pancreatic diseases (including PBT and CP). We excluded the healthy samples in Figure S7 because they were collected from health check-up center, where CA19-9 was not tested.
For example, in Figure 3a, there were 432 samples, including 169 pancreatic cancer cases, 78 PBT (with CA19-9 missing in 1 sample), 25 CP (with CA19-9 missing in 1 sample), and 160 healthy individuals (with all missing CA19-9). As a result, only 270 samples were available for analysis in Figure S7a. This difference in sample size explains the variation in the AUCs of the comb model between the two figures.

PCM score in supplementary table 10 is wrong. It must be a multiple of 0.5. PCM score for healthy samples are missing.
Response: Thanks for pointing out our mistake. Because an error occurred when exporting data from the compute. We have corrected this mistake in our revised version of manuscript, and this mistake will not affect the results.

The authors use serum and plasma interchangeably it seems. These are very different in the liquid biopsy field, as some assays work on one but not the other. Please clarify.
Response: We apologize for the error in our wording. It should have been “plasma” instead of serum. We have made the correction in the revised version of our manuscript.

There are grammatical errors that could benefit from close copy editing.
Response: We apologize for any confusion and grammar mistakes. We have carefully edited the manuscript in the revised version. We really hope that the language level has been substantially improved.

Reviewer #3 (Remarks to the Author):

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review *and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.*

Reviewer #4 (Remarks to the Author):

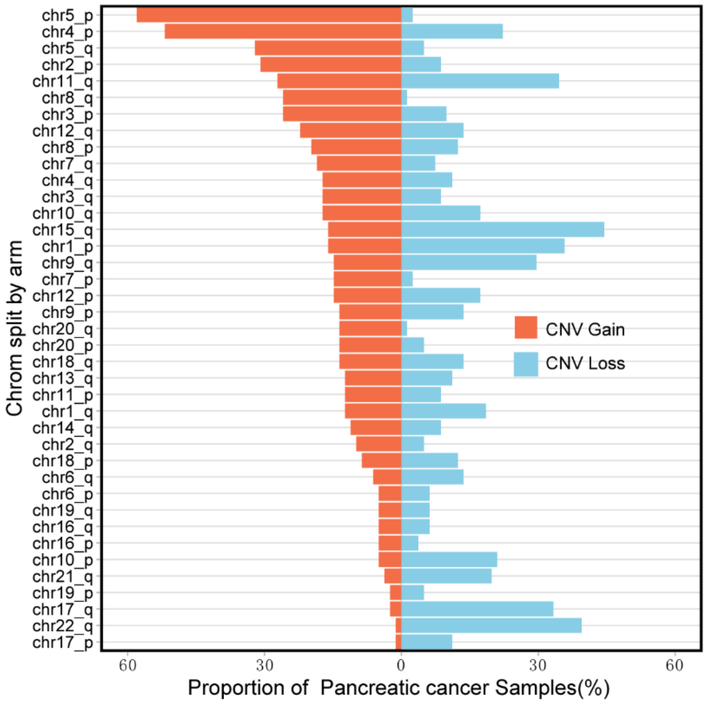
Wu and colleagues introduce a new approach for the earlier detection and prognosis of pancreatic cancer. The approach is based on a large number of samples (n=975) and uses clearly delineated training, validation and external cohorts consisting of pancreatic cancer, patients with benign tumors, chronic pancreatitis and healthy controls. In addition to using cell free DNA for their approach the authors assess other common biomarkers (CA19-9 and CEA). I greatly enjoyed reading the manuscript and think it is an advance that should be published. I think some edits would greatly improve the readability and clarify the results presented.

Response: Thank you for reading our manuscript. In the revised version of manuscript, we have made detailed modifications based on your feedback, and we hope it addresses all of your concerns.

Major:

- *In Figure 1A the samples are illustrated on the chromosome arm level but Figure 1E the rankings are based on chromosomes. Could you split the ranking by arm instead of chromosome and could you indicate whether the copy number alteration was a gain or a loss. Arm level events are more relevant in cancer than the entire chromosome. Also, some arms the events are opposing directions (ie 8p loss and 8q gain).*

Response: Thank you for your suggestion. We had not previously considered CNVs at the chromosome arm level. Based on your feedback, we have divided the chromosomes into different arms and created a figure to illustrate the gains and losses of CNVs on various chromosome arms in pancreatic cancer patients. As shown below, our analysis revealed that copy number alterations at the arm level included both gains and losses. Notably, certain chromosome arms, such as chr5p and chr8q, exhibited a predominant overall gain, while others, including chr17q, chr22q, and chr17p, showed a predominant overall loss.



- *According to The Cancer Genome Atlas (taken from fire browse and image reproduced below) the most common aneuploidies in pancreatic cancer are: 1q gain (28%), chr6p loss (36%), chr6q (46%), 7p gain (29%), 7q gain (27%), 8q gain (25%), 9p loss (42%), 9q loss (25%), 20q gain (25%). Could you explain why your 3 top features (chr5, chr4, and chr15) are not typically observed in pancreatic cancer? Also, could you explain why chr6, chr20 and chr7 are very common in pancreatic cancer (as assessed by the TCGA) but at the bottom of your rankings?*

Response: The differences in our results may stem from variations in ethnicity and sample detection methods. Additionally, discrepancies in sample sources could also contribute to variations in the findings, as our study utilized plasma cfDNA, whereas the DNA in the TCGA database is derived from tissue samples. Furthermore, based on a study publish in *Nature Communications* in 2023, CNVs in pancreatic tumor samples exhibit high heterogeneity, with varying degrees of CNV accumulation across different patients and tissue types [11]. Thus, we believe this could explain why our results are somewhat inconsistent with those in TCGA.

- *Please add the copy number score to Figure 3 and Supp Table 10.*

Response: Thanks for your suggestion, we have added the copy number score to Supp Table 10, since CNV score is a logistic variable, it is not well-suited for generating an ROC curve. To demonstrate the effectiveness of CNV in distinguishing between various groups, we have instead provided the positive ratios for different groups and stages of pancreatic cancer in Table S1 of the original manuscript.

- *"According to the distribution of the Zscore of healthy people in the region defined* *that the Zscore greater than 2 or less than -2 was the baseline threshold" (lines 153-154). Please clarify in the main text which cohort of healthy samples were used at this step. Isn't this a circular analysis that guarantees the specificity is 100%? This would also overestimate specificity if you were setting the copy number thresholds based on healthy samples prior to inclusion in the model.*

Response: The cohort we used to define the baseline of Zscore was training cohort, and all feature selection processes and threshold definitions were completed within this cohort. Therefore, this approach does not overestimate specificity. We have included this information in the main text (line 371 - line 373).

Minor:

• I don't really understand this sentence (lines 106-108): "Here, we performed a multi-center, large scale cohort study and employed a state-of-the art- next generation sequencing (NGS) technology to acquire plasma cfDNA end motif, nucleosome footprint, and fragments profiles on cfDNA from all enrolled cases." Why isn't copy number listed here. It is clearly a major part of the study and seems to separate out the cases and controls.

Response: Thanks for bring us to our attention, and we apologize for the mistake, the copy number was a major part of the study. The accurate phrasing should be: "Here, we performed a multi-center, large scale cohort study and employed state-of-the-art next generation sequencing (NGS) technology to acquire plasma cfDNA end motif, nucleosome footprint, fragments, and copy number variation (CNV) profiles on cfDNA from all enrolled cases."

We have made this correction in the main text. (line 97 - line 100)

• "There were a lot of studies working on the early detection of pancreatic cancer, for example, the extracellular vesicle long RNA, and exosomsomal micro RNAs" Seems like an incomplete list. There are other manuscripts that have used copy number, mutations, methylation, and fragmentomics to identify pancreatic cancer in liquid biopsies.

Response: Thank you for your suggestions. We have reviewed additional literature and compared it with our research, we have discussed these comparisons in the **Discussion** section.

Fragmentomes:

There was few research using fragmentomes to detect pancreatic cancer, but there was research focusing on liver cancer. Zachariah H. Foda, et.al used cfDNA fragmentome features, achieved high specificity in detecting liver cancer [12].

Copy number

CNV is generally not used alone as a diagnostic feature for cancer due to its low sensitivity. But there were research focusing on CNV-based prognostic model. For example, Qian Zhan et al. using CNV to classify pancreatic cancer patients, and a prognostic model was established [13].

Mutations

Zill et al. conducted a prospective analysis of five genes (KRAS, TP53, APC, FBXW7, and SMAD4) in tumor tissues and ctDNA from 26 pancreatic cancer patients. Among the 17 patients with successful sequencing, 90.3% (95% CI 73.1%–97.5%) of the mutations detected in tumor tissue were also identified in ctDNA. The diagnostic accuracy of ctDNA sequencing was 97.7%, with an average sensitivity of 92.3% and a specificity of 100% for the five genes [14].

Methylation

There were plenty of studies based on methylation of cfDNA, for example, Guler et al. used cfDNA 5-hydroxymethylcytosine features to detect early-stage pancreatic cancer, achieving strong performance (AUC > 0.9) in both discovery and independent test sets [15]. Ying, Lee et al. using 4-gene methylation panel (ADAMTS1, BNC1, LRFN5, and PXDN), achieving a sensitivity of 100% and specificity of 90% [16].

Others:

Based on extracellular vesicles long RNA profiling, Shulin Yu et al. developed a d-signature model for PDAC detection, the d-signature was able to identify early stage of pancreatic cancer (stage I/II) with an AUC of 0.949 [17].

Compared with previous studies, our study has the following advantage: 1. While others often focus on PDAC in their research, we have collected some pancreatic benign tumor, and we not only detect pancreatic cancer but also differentiate between cancer and non-cancer cases. 2. We validate our model in multicenter cohort; 3. PCP score is associated with overall survival, allowing for prognostic prediction; 4. Our study utilized four different types of features of cfDNA, allowing for a more comprehensive reflection of the differences in cfDNA among different populations.

- *Why are there missing scores in the PCP.Score (Column 18) in the Supp Table 10? Also, why are some of the missing scores "NA" but others "#N/A"?*

Response: The samples without a PCP score were excluded due to missing prognostic information, which led to their omission from the training of the prognostic model. Similarly, because these samples lacked prognostic information, we were unable to validate the prognostic model in the test and validation cohorts. Consequently, we did not calculate the PCP score for these samples.

For your second question, we have standardized all missing values as "NA".

- *I think on line 316 it should be prospective not perspective.*

Response: Thank you for pointing out our mistake. We have made the corrections in the manuscript.

Reference

1. Snyder MW, Kircher M, Hill AJ et al. Cell-free DNA Comprises an In Vivo Nucleosome Footprint that Informs Its Tissues-Of-Origin. Cell 2016; 164: 57-68.

2. Ulz P, Thallinger GG, Auer M et al. Inferring expressed genes by whole-genome sequencing of plasma DNA. Nat Genet 2016; 48: 1273-1278.

3. Davoli T, Xu AW, Mengwasser KE et al. Cumulative haploinsufficiency and triplosensitivity drive aneuploidy patterns and shape the cancer genome. Cell 2013; 155: 948-962.
4. Imperiale TF, Ransohoff DF, Itzkowitz SH et al. Multitarget stool DNA testing for colorectal-cancer screening. N Engl J Med 2014; 370: 1287-1297.
5. Chen L, Wu T, Fan R et al. Cell-free DNA testing for early hepatocellular carcinoma surveillance. EBioMedicine 2024; 100: 104962.
6. Chen L, Abou-Alfa GK, Zheng B et al. Genome-scale profiling of circulating cell-free DNA signatures for early detection of hepatocellular carcinoma in cirrhotic patients. Cell Res 2021; 31: 589-592.
7. Mouliere F, Chandrananda D, Piskorz AM et al. Enhanced detection of circulating tumor DNA by fragment size analysis. Sci Transl Med 2018; 10.
8. Cristiano S, Leal A, Phallen J et al. Genome-wide cell-free DNA fragmentation in patients with cancer. Nature 2019; 570: 385-389.
9. Jiang P, Sun K, Tong YK et al. Preferred end coordinates and somatic variants as signatures of circulating tumor DNA associated with hepatocellular carcinoma. Proc Natl Acad Sci U S A 2018; 115: E10925-E10933.
10. Jiang P, Sun K, Peng W et al. Plasma DNA End-Motif Profiling as a Fragmentomic Marker in Cancer, Pregnancy, and Transplantation. Cancer Discov 2020; 10: 664-673.
11. Zhang S, Fang W, Zhou S et al. Single cell transcriptomic analyses implicate an immunosuppressive tumor microenvironment in pancreatic cancer liver metastasis. Nat Commun 2023; 14: 5123.
12. Foda ZH, Annapragada AV, Boyapati K et al. Detecting Liver Cancer Using Cell-Free DNA Fragmentomes. Cancer Discov 2023; 13: 616-631.
13. Zhan Q, Wen C, Zhao Y et al. Identification of copy number variation-driven molecular subtypes informative for prognosis and treatment in pancreatic adenocarcinoma of a Chinese cohort. EBioMedicine 2021; 74: 103716.
14. Zill OA, Greene C, Sebisanovic D et al. Cell-Free DNA Next-Generation Sequencing in Pancreatobiliary Carcinomas. Cancer Discov 2015; 5: 1040-1048.
15. Guler GD, Ning Y, Ku CJ et al. Detection of early stage pancreatic cancer using 5-hydroxymethylcytosine signatures in circulating cell free DNA. Nat Commun 2020; 11: 5270.
16. Ying L, Sharma A, Chhoda A et al. Methylation-based Cell-free DNA Signature for Early Detection of Pancreatic Cancer. Pancreas 2021; 50: 1267-1273.
17. Yu S, Li Y, Liao Z et al. Plasma extracellular vesicle long RNA profiling identifies a diagnostic signature for the detection of pancreatic ductal adenocarcinoma. Gut 2020; 69: 540-550.

Reviewer #1 (Remarks to the Author):

The revision is very shoddy and perfunctory. In addition, as this work is soly based on previous biomarkers developed by others, the novelty could only be justified by a high-performance model. However, both the feature selection and building of the model are not convincing, with many technical concerns.

Response: Thank you for your thorough reading of our article and your detailed suggestions. Your feedback will greatly enhance the quality of our work.

1. The github page the authors provided does not exist with a 404 error.

Response: The code link is https://github.com/JimmyWu2024/PAAD_pipline. In fact, we discovered the error in the link and provided the editor with the new code link in the first revision. The editor had included the new link in the "Response to Referees Letter: Code and Data Links for Reviewers" document.

2. For data sharing, why not upload the aligned BED format file which does not contain any patient-sensitvie data? (echo Reviewer 2)

Response: We have provided our data in following link (In fastq format):

(A) Pancreatic disease samples: https://db.cngb.org/search/project/CNP0005903/,
(B) Healthy samples: https://db.cngb.org/search/project/CNP0006304/.

Additionally, for your convenience, we have provided the all of the raw data with below download link, which can be downloaded freely.

(A) Pancreatic disease samples: https://pan.baidu.com/s/1LunlyJrpmaKPJHRTM8eWMA?

(B) Healthy samples: https://pan.baidu.com/s/1cYERcvjCL8mEjnBTL1Ih-w?

Our raw data is in FASTQ format, if you want to get bed file, the processing is using bwa software to perform sequence alignment on the FASTQ file to obtain a BAM file, Then, in Linux system, you can use `samtools sort -n` to sort the BAM file, and subsequently convert it to BED format using `bedtools bamtobed`. The specific command for this step is:

```
bedtools bamtobed -i input.bam -bedpe > output.bed
```

3. You said you used "featureCounts" v2.19.1. This is funny, because the latest version of this tool is only v2.0.8. I do not understand how did you get the v2.19.1 version (from the future)?

Response: Thank you for pointing out our mistake, we indeed mistake for the version of featureCounts and Bedtools. In fact, the version of featureCounts was v1.6.2, the version of Bedtools was v2.19.1. We have corrected this mistake in our revised version of manuscript.

4. in "Filter Genes and Model Construction", did you do this step before model building? If so, there should be a disclosure of information to the testing dataset, which is inappropriate.

Response: Thank you for your attention to this issue. Maybe we didn't explain this step clearly in the text. In fact, all gene filtering and selection were carried out within the **training cohort**. The selected features were used for model construction, and the model was ultimately validated on independent testing cohort and validation cohorts. So, there won't be a disclosure of information to the testing dataset. We have revised this part of text in the manuscript (line 376).

5. "we considered the direction of CNVs relative to oncogenes and tumor suppressor genes to minimize false positives". Do you mean oncogenes should be amplified and suppressors should be LOH? This is not true, and definition of oncogenes/suppressors are not always correct. For example, in HCC, 1p is commonly LOH and 1q is commonly copy number gain, but samples with opposite trends are also found (Jiang et al. PNAS 2015).

Response: Thank you for your attention to this issue. The phenomenon you mentioned was exist. Actually, our method for calculating CNV score and defining oncogenes and tumor suppressor genes was derived from the paper published in 2013 *Cell* [1]. This is a classic study and the paper has been cited over 600 times. In this paper, the authors developed "TUSON explorer" to determine whether a gene is an oncogene or a tumor suppressor gene. In their evaluating system, gene amplification and deletion as one of indicator for identifying oncogenes and tumor suppressor genes. For tumor suppressor gene (TSG), the paper states: "the most predictive parameters for TSGs are: …high-level deletion frequency". Similarly, for oncogene (OG), it states: "the most predictive parameters for OGs are: …high-level amplification frequency". In our research, we used "TUSON explorer" to evaluate every gene, the results were shown in our code link: https://github.com/JimmyWu2024/PAAD_pipline(located in the CNV folder, with final results in Probability_TUSON.txt).

6. You use a panel of healthy subjects to model the copy number for others. In this case, you cannot use them anywhere during downstream model building. In Fig. 1e, the CNV signal is empty for all healthy subjects, I guess this should be due to your inappropriate re-analyzing of these cases. In fact, the signal cannot be so clean.

Response: Thanks for your concern. Actually, the healthy individuals used for modeling copy number come from the training cohort. We believe this will not impact other cohorts, as the training cohort is independent of the other cohorts.

The threshold was set using the CNV results from healthy individuals in the training cohort. When setting the threshold, the principle was that no large-scale copy number variations should be present on any chromosome of healthy individuals, ensuring that the specificity of the CNV indicator is 100%. This is also the reason why no CNV signals are shown for healthy individuals in the figure, as those not exceeding the threshold were not reported.

7. "Nucleosome footprint model was built by bedtools (https://bedtools.readthedocs.io/en/latest/) to calculate the reads distribution and remove uncovered genes at the TSS +/- 2.5kb region." This sentence appears twice in your response letter. Similarly, the WGS data processing part appears twice in your revised manuscript.

Response: Thank you for pointing out our mistake, we have corrected this in our revised version of manuscript.

8. In "procedure of GC correction", I have never seen such method based on your formula, most people use LASSO. Or, why donot use ichorCNA's normalization?

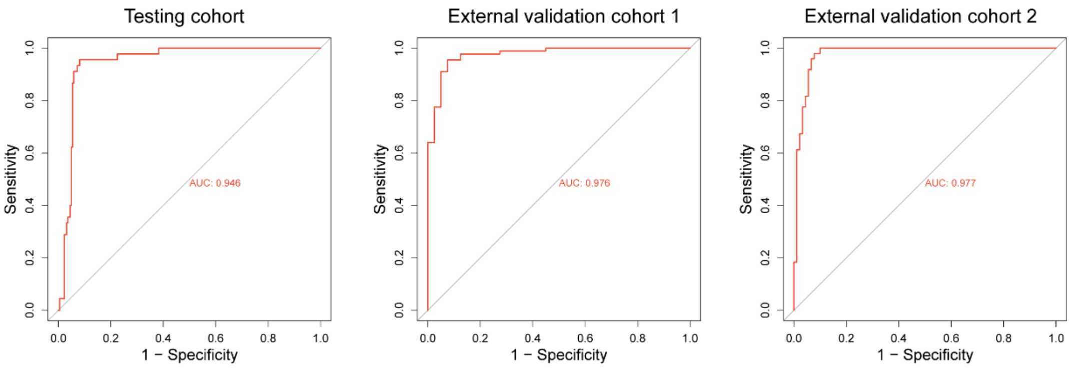
Response: Thank you for raising the question from a professional perspective. Actually, our method was referring to research published in 2008 *PNAS* [2]. We use a method that calculates the average coverage for each GC content bin, then normalizes it using the average coverage across all bins. For low-depth whole-genome sequencing data, we believe this approach is simpler and more effective. The reason we didn't use high-depth sequencing is due to considerations of sequencing cost and the potential for widespread adoption. While high-depth sequencing is more accurate, it is also more costly and complicated in terms of operation, which makes it less suitable for large-scale implementation.

9. "The formula for the single score indicates that at least two out of three dimensions must exceed their respective thresholds to reach the positive threshold of the final PCM score. This approach is intended to prevent a low score in one dimension from negatively impacting the overall logistic score." Is it just bagging? What is the advantage over GBDT?

Response: Yes, we used bagging method, it is a parallel training method, it has lower computation cost and it is more suitable in large sample, and bagging method was used to train multiple independent models.

GBDT is a boosting method, it has higher computation cost, boosting was used to train models sequentially, which means the models were dependent from each other, but for our data, we are not sure whether there is a correlation between these indicators.

Additionally, we also used the GBDT model to integrate motif, NF, fragment, and CNV, and obtained a prediction model in R software, using ‘gbm’ package (version 2.1.8.1). However, the performance of this model was not as good as our bagging model. The ROC curve and AUC value was shown below. We evaluate the performance of GBDT model in classification of pancreatic cancer to non-cancer groups in Testing cohort, External validation cohort 1, and External validation cohort 2. We can see that, the AUC value of GBDT model in Testing cohort, External validation cohort 1 and External validation cohort 2 was 0.946, 0.976, and 0.977. While in our bagging model, the AUC value was 0.979, 0.992, and 0.986. The performance of bagging model and GBDT model is similar in all three cohorts. Thus, we think bagging is also suitable for our data.



10. In "Feature selection", you just removed chrY, which means you kept chrX? If so, how do you handle the dosage differences in male and female samples on chrX?

Response: Yes, we removed chrY and kept chrX, we applied different handling methods for different types of features. For CNV, the baseline for the female X chromosome is two copies, while the baseline for the male X chromosome is one copy during statistical analysis; For NF, the data for the male X chromosome will be corrected based on the proportion of the female X chromosome data (statistically, the proportion of data from the X chromosome in females accounts for approximately 4.8% of the total data). The Fragment and Motif are both converted into ratios during feature quantification, such as the ratio of long to short fragment counts or the proportion of a specific motif type. After this conversion, they might not be affected by gender bias.

11. "Procedure for quantifying end motif (based on methods in 2018 PNAS, and 2020 Cancer Discovery [9, 10])", there is NO 2018 PNAS paper, and your citation 9 is also incorrect. There are also other incorrect citations in the manuscript, e.g., citations 9 and 31. This made me doubt the familiarity to the field of the authors.

Response: In the first version of response to referees, if you check it, you will find the citation 9 is : Jiang P, Sun K, Tong YK et al. Preferred end coordinates and somatic variants as signatures of circulating tumor DNA associated with hepatocellular carcinoma. Proc Natl Acad Sci U S A 2018; 115: E10925-E10933, and it is actually a paper published in 2018 in *PNAS*. In this paper, the authors have demonstrated the presence of tumor associated cell-free DNA preferred end coordinates among HCC patients, which provides insights for us to explore cfDNA end sequence characteristics in pancreatic cancer. Thus, we think we should cite this article.

We might perhaps we placed the superscript in the wrong position, which led to your misunderstanding. The right citation is:

Procedure for quantifying end motif (based on methods in 2018 PNAS [9], and 2020 Cancer Discovery [10].

Reference

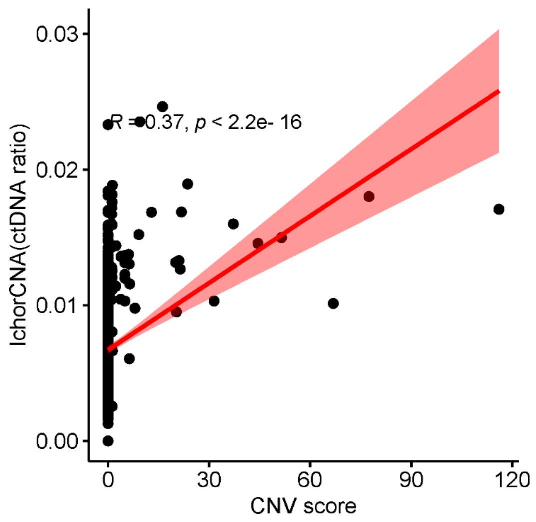
9. Jiang P, Sun K, Tong YK et al. Preferred end coordinates and somatic variants as signatures of circulating tumor DNA associated with hepatocellular carcinoma. Proc Natl Acad Sci U S A 2018; 115: E10925-E10933.

10. Jiang P, Sun K, Peng W et al. Plasma DNA End-Motif Profiling as a Fragmentomic Marker in Cancer, Pregnancy, and Transplantation. Cancer Discov 2020; 10: 664-673.

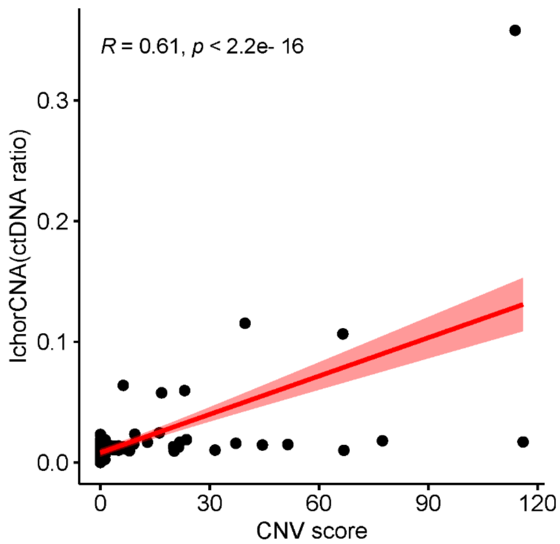
Regarding the issue you mentioned with incorrect citations in the article, we have rechecked the references and found that there were indeed some errors. We have corrected these citations in the revised manuscript. Thank you for pointing out our mistakes.

12. You show that your CNV score is correlated with ichorCNA. However, you can see that the regression line is biased by the samples with high tumor DNA fraction. You should use log-scaled values, or focus on those with 5% or less ichorCNA scores, as these are more important early-stage samples.

Response: Thanks for your suggestion, we have focused on samples less than 5% ichorCNA scores. The correlation was shown below.



When we focused on those early-stage pancreatic cancer, the correlation was shown below. We can see that the correlation between CNV score and ichorCNA is still strong.



13. For the disconcordence between your CNA analysis and TCGA, your explaniation is that you are working on cfDNA while TCGA is working on tissue. This is definitely unacceptable, as CNAs inferred from plasma is known to be highly consistent with tumor tissues. Tumor heterogenety and patients' races could be more reasonable explainations.

Response: Thanks for your kindly suggestion. We admit that there was an issue with our way of phrasing this. Through literature research, we found that the genetic mutations in cfDNA are indeed highly consistent with those in tissue samples, which was found by research published in 2018 *Cancer Discovery*[3]. Therefore, the discrepancy between our CNV analysis and the TCGA data is probably due to tissue heterogeneity and racial differences.

14. Typos, e.g., "Tabel S3-S5" in line 423.

Response: Thank you for pointing out the errors in our article. We have made the corrections, and we have also carefully proofread the entire article.

Reviewer #1 (Remarks on code availability):

The github page is not available, and I cannot view the code.

Response: The code link is https://github.com/JimmyWu2024/PAAD_pipline. In fact, we discovered the error in the link and provided the editor with the new code link. The editor had included the new link in the "Response to Referees Letter: Code and Data Links for Reviewers" document.

Reviewer #2 (Remarks to the Author):

The authors did a good job of addressing my concerns. I am satisfied.

Response: Thank you for reviewing our manuscript and for your valuable feedback. We are pleased to hear that you are satisfied with the revised version.

Reviewer #3 (Remarks to the Author):

I co-reviewed this manuscript with one of the reviewers who provided the listed reports. This is part of the Nature Communications initiative to facilitate training in peer review and to provide appropriate recognition for Early Career Researchers who co-review manuscripts.

Response: Thank you for reviewing our manuscript and for your valuable feedback.

Reviewer #4 (Remarks to the Author):

• The provided list seems to ignore much of the published literature for the detection of pancreatic cancer in plasma.

Fragmentomes: There was few research using fragmentomes to detect pancreatic cancer, but there was research focusing on liver cancer. Zachariah H. Foda, et.al used cfDNA fragmentome features, achieved high specificity in detecting liver cancer [12]

• That is not the correct citation. This is the correct citation (Genome-wide cell-free DNA fragmentation in patients with cancer –Cristiano et al 2019). The Velculescu lab (DELFI group) looked at fragmentation patterns in pancreatic (n=34) cancers in their 2019 manuscript.

Response: Thank you for providing us with the reference. We indeed missed this paper. In this article, researchers from the Velculescu lab analyzed the fragmentation profiles of cfDNA from 236 cancer patients, including 34 pancreatic cancer patients, and established a cancer diagnostic model. The sensitivity of the model ranged from 57% to 99%, with a specificity of 98% [4]. We have now included this article in the discussion section of our manuscript (line227 - line230).

Copy number

CNV is generally not used alone as a diagnostic feature for cancer due to its low sensitivity. But there was research focusing on CNV-based prognostic model. For example, Qian Zhan et al. using CNV to classify pancreatic cancer patients, and a prognostic model was established [13].

• The Vogelstein group has done extensive work on aneuploidy/CNAs as a biomarker in pancreatic cancer (and other cancers). (Assessing aneuploidy with repetitive element sequencing Douville et al 2020). That manuscript evaluates plasma from patients with pancreatic cancer (n=82) and reports sensitivity for copy number analysis. They also did a follow-up study looking at copy number alterations and integrating it with alu sines and proteins in 287 plasma samples from patients with pancreatic cancer (Machine learning to detect the SINEs of cancer--Douville et al 2024).

Response: Thanks for you providing us with the reference. We indeed missed these papers. In 2020 *PNAS* [5], the Vogelstein group developed RealSeqS to detect aneuploidy/CNAs, which can detected various cancers (including pancreatic cancer), combining aneuploidy with somatic mutation and protein biomarkers can achieved median sensitivity of 80% of detecting cancers. In their follow-up research, they developed A-plus, which can enhanced sensitivity over that achieved for aneuploidy alone at matched specificities [6]. We have added these papers to the discussion section (line230 – line235). We believe that your suggestion will greatly enhance the quality of our article.

• "CNV is generally not used alone as a diagnostic feature for cancer due to its low sensitivity."—This statement is irrelevant to whether copy number alterations have been evaluated in pancreatic cancer or not. It is important to note that many liquid biopsies papers have suggested integrating multiple analytes due to enhance sensitivity. In fact, your manuscript is specially making the argument that you need to integrate end motif, nucleosome footprint, fragmentation, and copy number to garner sufficient sensitivity to detect pancreatic cancer.

Response: Your suggestion is very valuable to us. We acknowledge that there is an issue with our wording.

Mutations Zill et al. conducted a prospective analysis of five genes (KRAS, TP53, APC, FBXW7, and SMAD4) in tumor tissues and ctDNA from 26 pancreatic cancer patients. Among the 17 patients with successful sequencing, 90.3% (95% CI 73.1%–97.5%) of the mutations detected in tumor tissue were also identified in ctDNA. The diagnostic accuracy of ctDNA sequencing was 97.7%, with an average sensitivity of 92.3% and a specificity of 100% for the five genes [14].

• Cohen et al 2017 (Combined circulating tumor DNA and protein biomarker-based liquid biopsy for the earlier detection of pancreatic cancers) should be included because it is one of the seminal works on pancreas mutations in plasma. This manuscript evaluates mutations in plasma from 221 patients with pancreatic cancer.

Response: Your suggestion is very valuable to us. We have carefully reviewed Cohen's work, in their research, they used KRAS mutations combined with four protein biomarkers (CA19-9, CEA, HGF, OPN) achieved sensitivity of 64% in detecting PDAC. We have included this research in the discussion section of our revised manuscript. (line247 - line249)

Methylation:

• Grail's paper should be cited (Sensitive and specific multi-cancer detection and localization using methylation signatures in cell-free DNA—Liu et al 2020). They have done extensive analysis on detection (and also prognostication) of pancreatic cancers using methylation.

Response: Thank you for pointing out our shortcomings. We indeed missed this paper. In this paper, they enrolled more than 50 types of cancer (including pancreatic cancer), through using methylation signatures in cfDNA, achieved high sensitivity in detecting early stage of pancreatic cancer [7]. In their extensive work, they evaluated the prognostic significance of cancer [8]. And in their 2021 *Annals of Oncology* [9], and 2022 *Cancer cell* [10], they did extensive work on methylation-based pancreatic cancer detection. We have included the references in the discussion section of our manuscript. (line239 – line 242)

Minor:

• Copy Number Variation (CNV) refers to germline events. Copy Number Alteration (CNA) refers to somatic events. The terminology should be consistent with the field.

Response: Thanks for your suggestion, we have corrected this mistake in our manuscript.

References

1 Davoli, T. *et al.* Cumulative haploinsufficiency and triplosensitivity drive aneuploidy patterns and shape the cancer genome. *Cell* **155**, 948-962, doi:10.1016/j.cell.2013.10.011 (2013).

2 Fan, H. C., Blumenfeld, Y. J., Chitkara, U., Hudgins, L. & Quake, S. R. Noninvasive diagnosis of fetal aneuploidy by shotgun sequencing DNA from maternal blood. *Proc Natl Acad Sci U S A* **105**, 16266-16271, doi:10.1073/pnas.0808319105 (2008).

3 Strickler, J. H. *et al.* Genomic Landscape of Cell-Free DNA in Patients with Colorectal Cancer. *Cancer Discov* **8**, 164-173, doi:10.1158/2159-8290.CD-17-1009 (2018).

4 Cristiano, S. *et al.* Genome-wide cell-free DNA fragmentation in patients with cancer. *Nature* **570**, 385-389, doi:10.1038/s41586-019-1272-6 (2019).

5 Douville, C. *et al.* Assessing aneuploidy with repetitive element sequencing. *Proc Natl Acad Sci U S A* **117**, 4858-4863, doi:10.1073/pnas.1910041117 (2020).

6 Douville, C. *et al.* Machine learning to detect the SINEs of cancer. *Sci Transl Med* **16**, eadi3883, doi:10.1126/scitranslmed.adi3883 (2024).

7 Liu, M. C. *et al.* Sensitive and specific multi-cancer detection and localization using methylation signatures in cell-free DNA. *Ann Oncol* **31**, 745-759, doi:10.1016/j.annonc.2020.02.011 (2020).

8 Chen, X. *et al.* Prognostic Significance of Blood-Based Multi-cancer Detection in Plasma Cell-Free DNA. *Clin Cancer Res* **27**, 4221-4229, doi:10.1158/1078-0432.CCR-21-0417 (2021).

9 Klein, E. A. *et al.* Clinical validation of a targeted methylation-based multi-cancer early detection test using an independent validation set. *Ann Oncol* **32**, 1167-1177, doi:10.1016/j.annonc.2021.05.806 (2021).

10 Jamshidi, A. *et al.* Evaluation of cell-free DNA approaches for multi-cancer early detection. *Cancer Cell* **40**, 1537-1549 e1512, doi:10.1016/j.ccell.2022.10.022 (2022).

REVIEWERS' COMMENTS

Reviewer #1 (Remarks to the Author):

This version has improved a lot, but there are still many grammar errors in the response letter and main text that you need to fix.

Response: Thank you for your valuable suggestions. We have carefully checked the grammar in the main text and response letter and corrected the errors.

For the descriptions on Cristiano and Cohen studies, I suggest move them to Introduction rather than put in Discussion.

Response: Your suggestion is very reasonable. After carefully consideration, we have moved the Cristinano and Cohen's work along with others work in cfDNA-based pancreatic cancer detection to the Introduction section. We believe that including this information earlier in the manuscript will help readers better understand the background and context of our research.

The Cristiano dataset has been reanalyzed in various studies, inclduing An et al. Nature Communications 2023 and Ju et al. Cell Reports Methods 2024 with improved performance on pancreatic cancer. In addition, Bie et al. Nature Communications 2023 also investigated pancreatic cancer with high accuracy. You should discuss/cite these papers.

You should also provide the name of your ethics committee.

Response: Thanks for your valuable suggestion. We have reviewed the papers you provided. An et al. found that DNA methylation might regulate cfDNA fragmentation, they developed a cfDNA end-preference-based metric for cancer diagnosis[1]. Ju et al. investigated the cfDNA fragmentomic characteristics against nucleosome positioning patterns in hematopoietic cells and developed a cancer diagnostic model based on the cfDNA fragmentomic metrics[2]. Bie et al. adapt an enzyme-mediated methylation sequencing method and developed a genome-wide cfDNA methylation, fragmentation, and copy number alteration (CNA) characteristics integrated model for cancer detection[3]. We have incorporated the above content into the manuscript (line 209 - line 214, line 216 - line 218).

According to your suggestion, we provided the name of ethics committee in our manuscript (line 305 - line 307).

Reviewer #1 (Remarks on code availability):

May add more documentation.

Response: Thanks for your valuable suggestion. According to your suggestion, we have added incorporated more previous research findings to the manuscript.

Reviewer #4 (Remarks to the Author):

Authors have addressed my concerns.

Response: Thank you for reviewing our manuscript and for your valuable feedback. We are pleased to hear that you are satisfied with the revised version.

Reference

[1] An Y, Zhao X, Zhang Z, et al. DNA methylation analysis explores the molecular basis of plasma cell-free DNA fragmentation [J]. Nat Commun, 2023, 14(1): 287.

[2] Ju J, Zhao X, An Y, et al. Cell-free DNA end characteristics enable accurate and sensitive cancer diagnosis [J]. Cell Rep Methods, 2024, 4(10): 100877.

[3] Bie F, Wang Z, Li Y, et al. Multimodal analysis of cell-free DNA whole-methylome sequencing for cancer detection and localization [J]. Nat Commun, 2023, 14(1): 6042.

Wu and colleagues introduce a new approach for the earlier detection and prognosis of pancreatic cancer. The approach is based on a large number of samples (n=975) and uses clearly delineated training, validation and external cohorts consisting of pancreatic cancer, patients with benign tumors, chronic pancreatitis and healthy controls. In addition to using cell free DNA for their approach the authors assess other common biomarkers (CA19-9 and CEA). I greatly enjoyed reading the manuscript and think it is an advance that should be published. I think some edits would greatly improve the readability and clarify the results presented.

Major:

- In Figure 1A the samples are illustrated on the chromosome arm level but Figure 1E the rankings are based on chromosomes. Could you split the ranking by arm instead of chromosome and could you indicate whether the copy number alteration was a gain or a loss. Arm level events are more relevant in cancer than the entire chromosome. Also, some arms the events are opposing directions (ie 8p loss and 8q gain).
- According to The Cancer Genome Atlas (taken from fire browse and image reproduced below) the most common aneuploidies in pancreatic cancer are: 1q gain (28%), chr6p loss (36%), chr6q (46%), 7p gain (29%), 7q gain (27%), 8q gain (25%), 9p loss (42%), 9q loss (25%), 20q gain (25%). Could you explain why your 3 top features (chr5, chr4, and chr15) are not typically observed in pancreatic cancer? Also, could you explain why chr6, chr20 and chr7 are very common in pancreatic cancer (as assessed by the TCGA) but at the bottom of your rankings?
- Please add the copy number score to Figure 3 and Supp Table 10.
- “According to the distribution of the Zscore of healthy people in the region defined that the Zscore greater than 2 or less than -2 was the baseline threshold” (lines 153-154). Please clarify in the main text which cohort of healthy samples were used at this step. Isn’t this a circular analysis that guarantees the specificity is 100%? This would also overestimate specificity if you were setting the copy number thresholds based on healthy samples prior to inclusion in the model.

gdac.broadinstitute.org/runs/analyses__2016_01_28/reports/cancer/PAAD-TP/CopyNumber_Gistic2/nozzle.html

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Table 3. Arm-level significance table - 21 significant results found. The significance cutoff is at Q value=0.25.

Arm	# Genes	Amp Frequency	Amp Z score	Amp Q value	Del Frequency	Del Z score	Del Q value
1p	1300	0.10	-0.299	1	0.25	5.95	7.71e-09
1q	1195	0.28	6.91	9.64e-11	0.09	-0.895	1
2p	624	0.09	-2.21	1	0.07	-2.98	1
2q	967	0.10	-1.05	1	0.02	-3.95	1
3p	644	0.05	-3.09	1	0.22	2.68	0.0126
3q	733	0.12	-0.878	1	0.12	-0.683	1
4p	289	0.07	-3.27	1	0.14	-0.75	1
4q	670	0.04	-3.55	1	0.15	0.135	1
5p	183	0.16	-0.419	1	0.06	-3.78	1
5q	905	0.11	-0.891	1	0.09	-1.5	1
6p	710	0.06	-2.6	1	0.36	8.43	0
6q	556	0.04	-3.11	1	0.46	11.7	0
7p	389	0.29	4.68	1.97e-05	0.02	-4.3	1
7q	783	0.27	5.08	3.95e-06	0.03	-3.64	1
8p	338	0.15	-0.339	1	0.28	4.13	6.82e-05
8q	551	0.25	3.7	0.00109	0.16	0.352	0.874
9p	301	0.07	-2.61	1	0.42	9.13	0
9q	700	0.08	-2.14	1	0.25	4.25	4.29e-05
10p	253	0.07	-3.26	1	0.21	1.38	0.221
10q	738	0.08	-1.97	1	0.18	1.36	0.221
11p	509	0.08	-2.67	1	0.11	-1.73	1
11q	975	0.09	-1.39	1	0.10	-0.974	1
12p	339	0.17	0.383	1	0.15	-0.519	1
12q	904	0.15	0.902	0.836	0.17	1.71	0.137
13q	560	0.17	0.751	0.927	0.13	-0.756	1
14q	938	0.15	1.02	0.785	0.09	-1.23	1
15q	810	0.10	-1.37	1	0.18	1.61	0.156
16p	559	0.13	-0.836	1	0.07	-2.92	1
16q	455	0.13	-0.773	1	0.08	-2.63	1
17p	415	0.08	-2.24	1	0.46	11	0
17q	972	0.12	-0.405	1	0.12	-0.405	1
18p	104	0.19	0.452	1	0.37	6.37	6.26e-10
18q	275	0.11	-1.39	1	0.62	16.1	0
19p	681	0.09	-1.85	1	0.12	-0.685	1
19q	935	0.17	1.57	0.397	0.07	-2.12	1
20p	234	0.20	1.18	0.696	0.15	-0.762	1
20q	448	0.25	3.54	0.00163	0.04	-3.67	1
21q	258	0.02	-4.56	1	0.32	5.46	1.22e-07
22q	564	0.05	-3.23	1	0.27	4.33	3.45e-05
Xp	418	0.04	-3.88	1	0.16	0.00209	1
Xq	668	0.07	-2.59	1	0.11	-1.23	1

Minor:

- I don't really understand this sentence (lines 106-108): "Here, we performed a multi-center, large scale cohort study and employed a state-of-the art- next generation sequencing (NGS) technology to acquire plasma cfDNA end motif, nucleosome footprint, and fragments profiles on cfDNA from all enrolled cases." Why isn't copy number listed here. It is clearly a major part of the study and seems to separate out the cases and controls.
- "There were a lot of studies working on the early detection of pancreatic cancer, for example, the exracellular vesicle long RNA, and exosomsomal micro RNAs"→ Seems like an incomplete list. There are other manuscripts that have used copy number, mutations, methylation, and fragmentomics to identify pancreatic cancer in liquid biopsies.
- Why are there missing scores in the PCP.Score (Column 18) in the Supp Table 10? Also, why are some of the missing scores "NA" but others "#N/A"?

- I think on line 316 it should be prospective not perspective.